



EUROPEAN
**MICROSOFT
FABRIC+SQL**
Community Conference

BARCELONA
28 SEPT - 01 OCT 2026





the ultimate decision guide

capacity strategies to optimise cost and compute

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**start with a plan,
then monitor to optimise.**

Fabric capacity & capacity based pricing



One capacity to power all your Fabric computes

Less need for scaling of components

One capacity = one price

Easier-to-read invoice

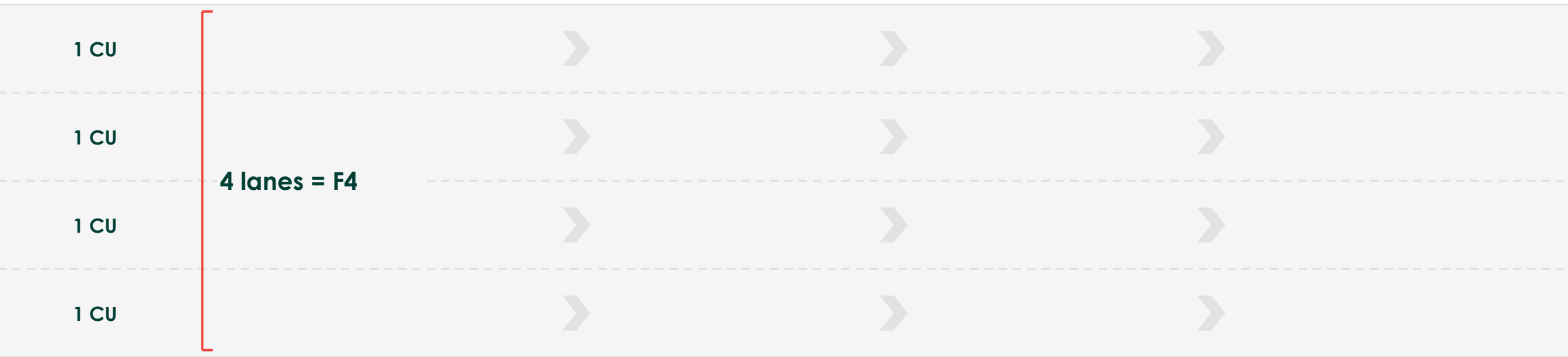
More predictable monthly costs

predictability over flexibility

Fabric capacity = highway

1 lane = 1 CU *

Current highway = F4



* Not really, for illustration purposes only

Two operation types



Background operations

Longer running, typically scheduled operations.

e.g. Data Factory pipeline, PySpark notebook, semantic model refresh



Interactive operations

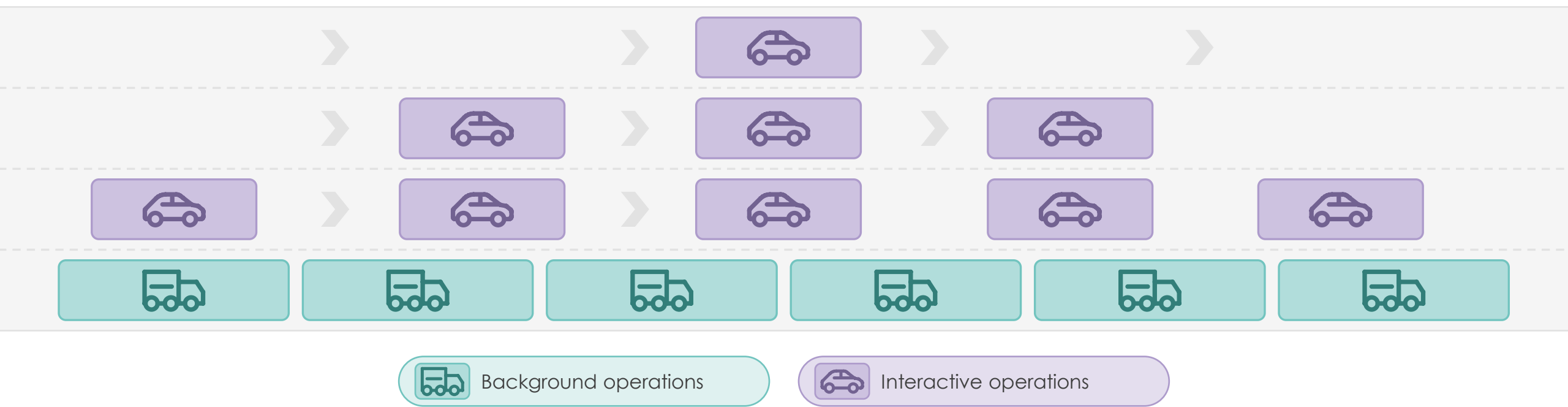
On-demand requests, triggered by someone using the UI.

e.g. Power BI render, DirectQuery, report visual queries

Background operations get priority over interactive operations.

Background & interactive operations

All Fabric operations fill your capacity



Bursting & smoothing

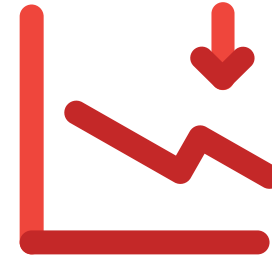
Bursting

Temporarily use more capacity than you bought.

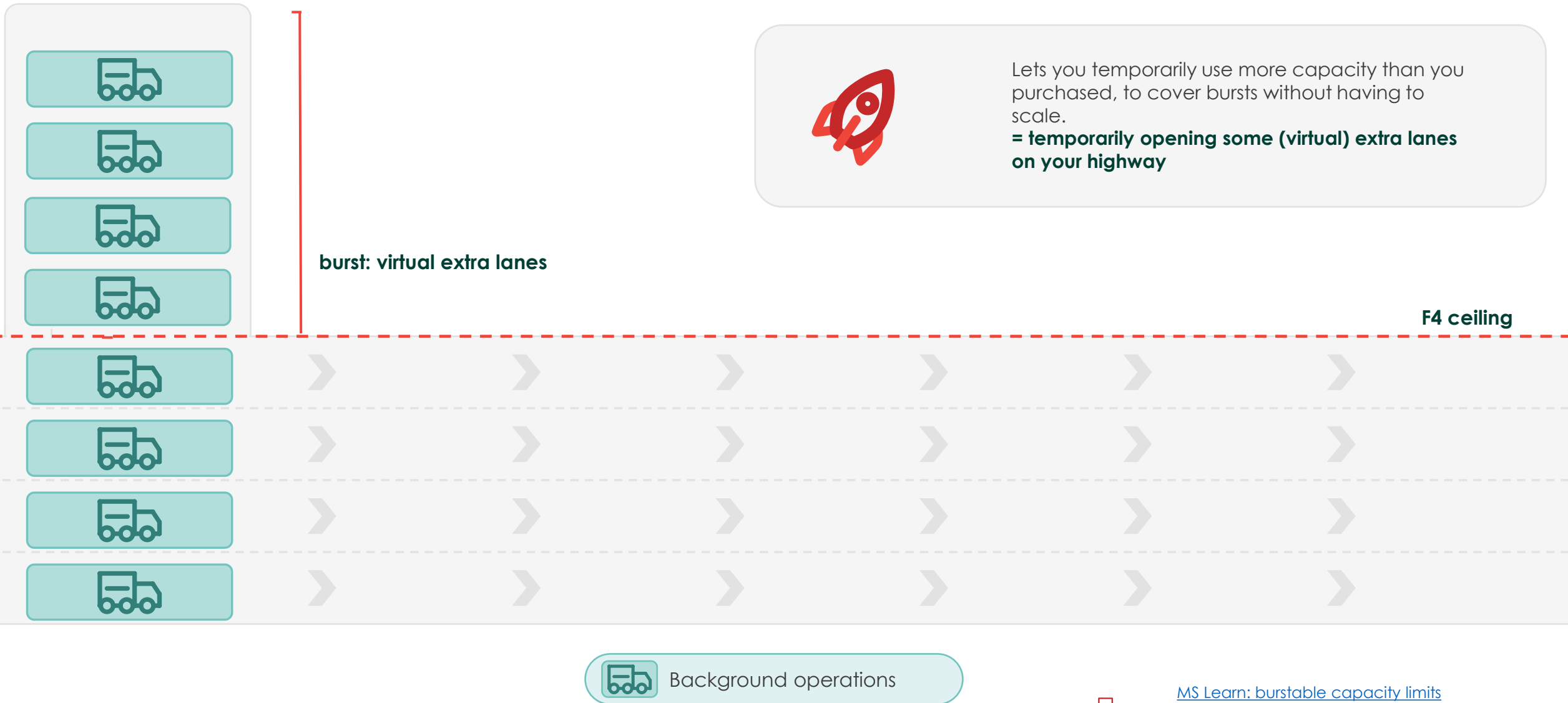


Smoothing

Spread what an operation costs over the next 24 hours.

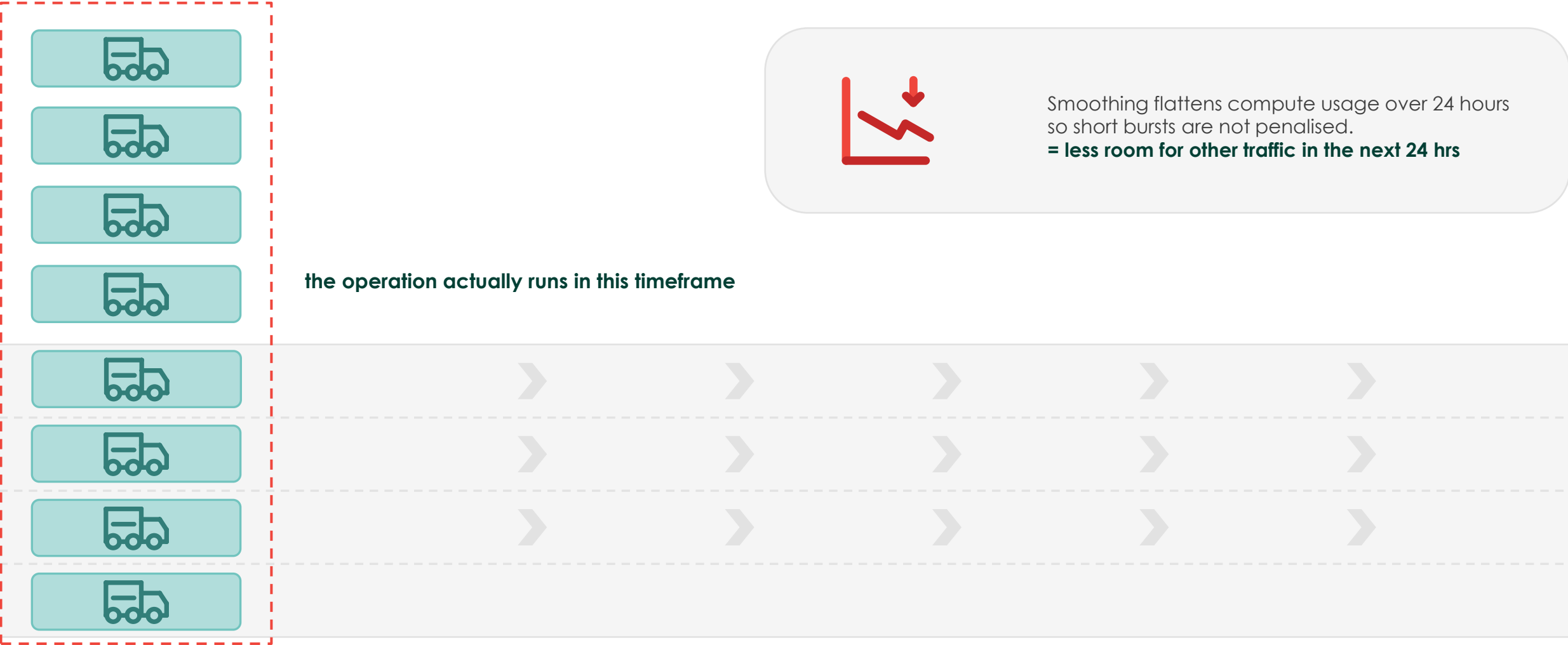


Bursting: temporarily open extra lanes



[MS Learn: burstable capacity limits](#)

Smoothing: flatten usage over 24 hours



Smoothed 24 h vs smoothed 5 min



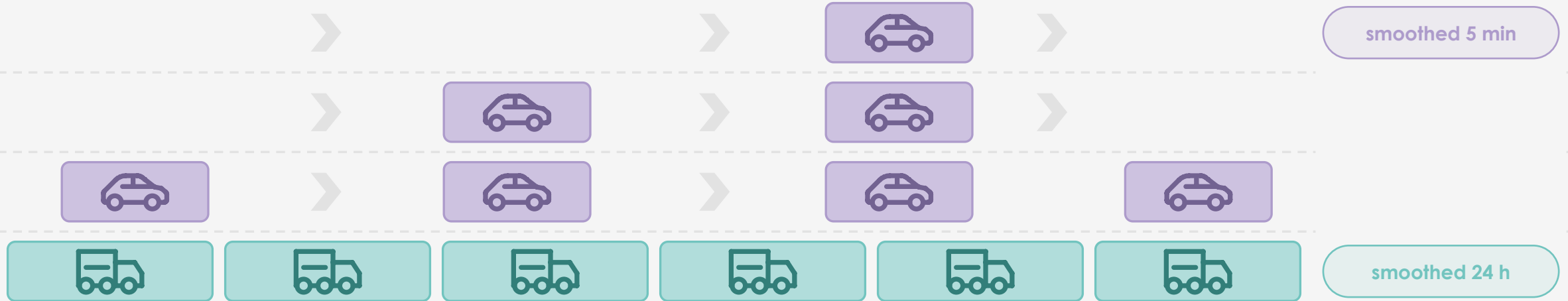
Background operations

Smoothed 24 h



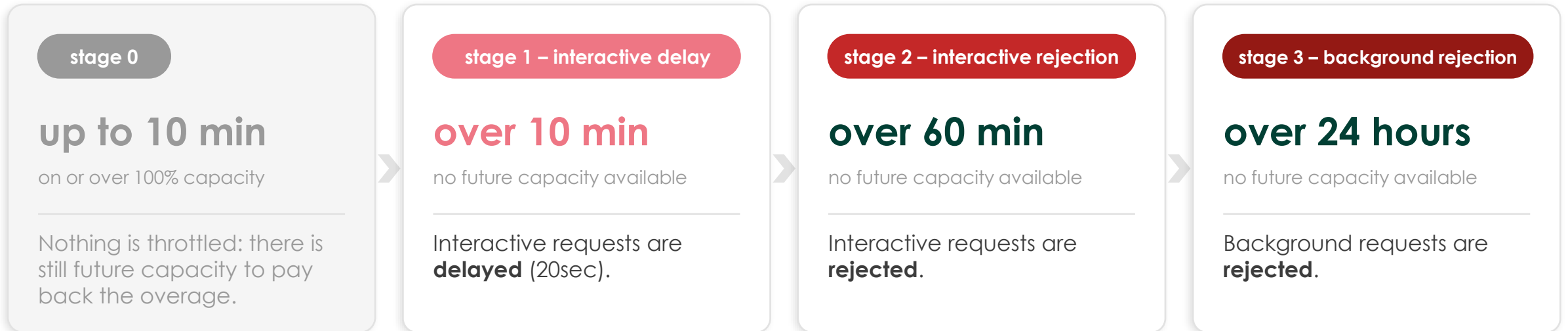
Interactive operations

Smoothed 5 min



At capacity?

A capacity limit is hit when there is no more future capacity to “pay back” the over usage (overage)



What draws CU

Consumption rates vary hugely across workload types

Background - smoothed 24h

-  Power BI: **scheduled refresh**, subscriptions, Direct Lake
-  Data Factory pipelines & **Dataflow Gen2**
-  Warehouse
-  Lakehouse
-  Notebook & Spark Job
-  OneLake
-  **Real-time analytics** (Eventstream)
-  Copilot

Interactive - smoothed 5 min

-  Power BI: render, DirectQuery
-  **Interactive queries to semantic model** & web modelling
-  KQL uptime
-  Fabric API for GraphQL
-  Fabric Database SQL queries
-  AI Functions in Power BI
-  User Data Functions

CU = billing rate x billed duration

plan: what capacity do we need?

Plan: what capacity do we need?

 [Fabric Capacity Estimator](#)

Microsoft Fabric SKU Estimator (preview)

This preview of the Microsoft Fabric SKU Estimator is made available under Microsoft's [terms of use](#). Estimates generated may not be accurate.

Tell us about your data and we'll generate a SKU recommendation based on your capacity requirements.

Data Information

Total size of the data when compressed (GiB) ⓘ

1000

Number of daily batch cycles ⓘ

1

Number of tables across all data sources ⓘ

250

Reset

Fabric usage

Select the workloads and features that you plan to use in Fabric. Some may require additional information.

☒ Ad-Hoc SQL Analytics ⓘ

☐ Data Factory ⓘ

☒ Data Warehouse ⓘ

☐ Spark Jobs ⓘ

☒ Data Agents ⓘ

☐ Data Science ⓘ

☐ Fabric Plan ⓘ



Estimation

Estimated Viable SKU

F128

DataAgent (6%)

SQL analytics endpoint (Less than 1%)

Data warehouse (7%)

Eventstream (21%)

Power BI (25%)

OneLake Operations (Less than 1%)

Unused (40%)

Storage

OneLake: 1700 Gb

Licenses

Power BI Pro: 20

Was the SKU Estimations helpful to you?

/ [Share feedback](#)

Start your free Microsoft Fabric trial now. [Learn more](#)



ONEUROPE

T-shirt sizes based on our projects

characteristic	S small	M medium	L large	XL very large
platform size	0.05 TB	0.5 TB	5 TB	50 TB
nr. of tables in the estate	~300	~2.000	~15.000	~75.000
largest transactional table row count	10M-100M	100M-1B	1B-10B	10B-100B
largest semantic model	0.5 GB	3.5 GB	18 GB	80 GB
viewers	30	250	700	2,500
creators	3	8	100	250
peak-hour share	40%	35%	30%	25%
avg. refreshes per day	2	4	24	24
pipeline hours	0.5	2	8	16
Suggested starting point	F4	F32	F128	F512
monthly capacity price, reserved 1yr	€ 283	€ 2.267	€ 9.067	€ 36.270

it's not an exact science – go with what ticks most boxes

Workload type

1



Data pipelines

driven by **how much data you move and how often you move it**, and billed on how long the engine runs rather than on volume.

2



Semantic models & power bi

driven by **model size, complexity and refresh rate**

3



Real-time intelligence

Driven by **how many eventstreams and eventhouses** you keep running, because it bills uptime rather than volume.

4



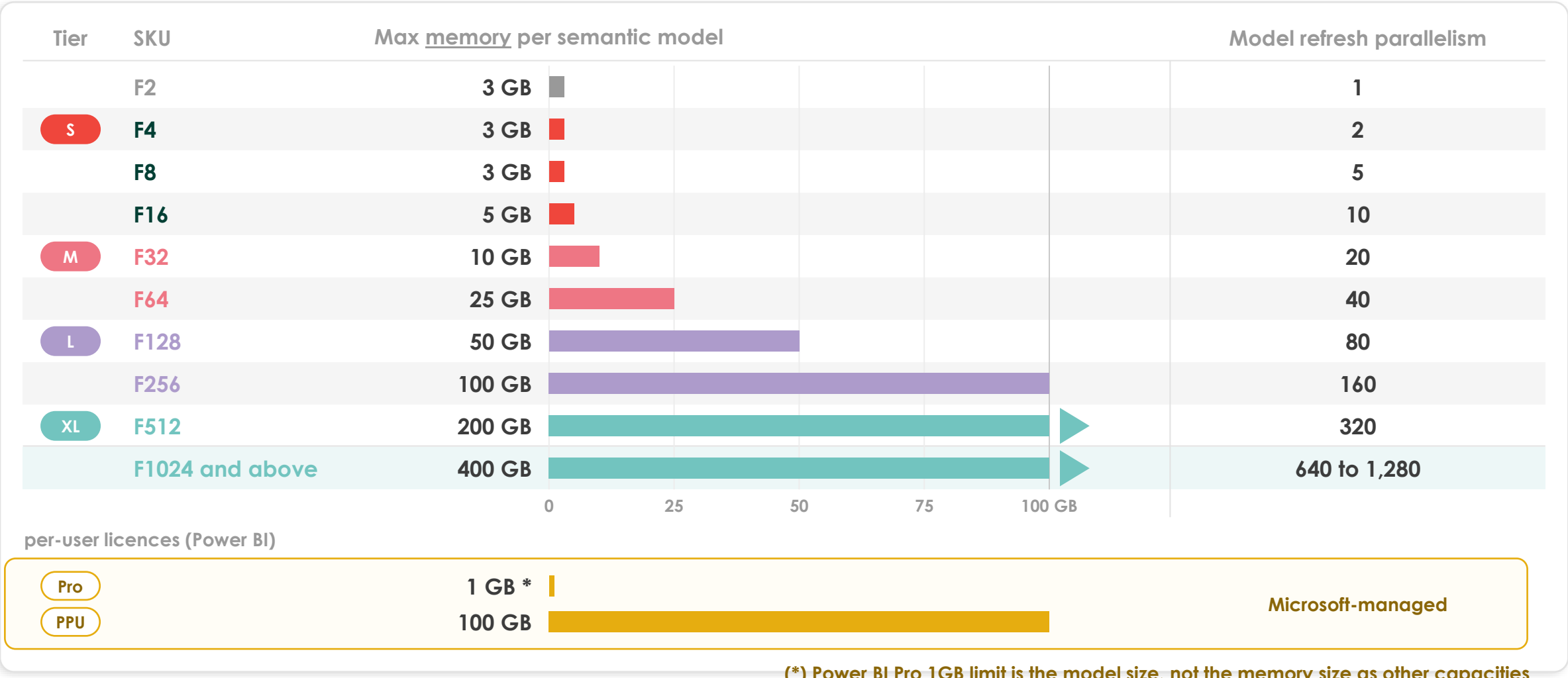
AI augmented

Genuinely hard to estimate, can peak hard

Fabric vs. power BI workspaces

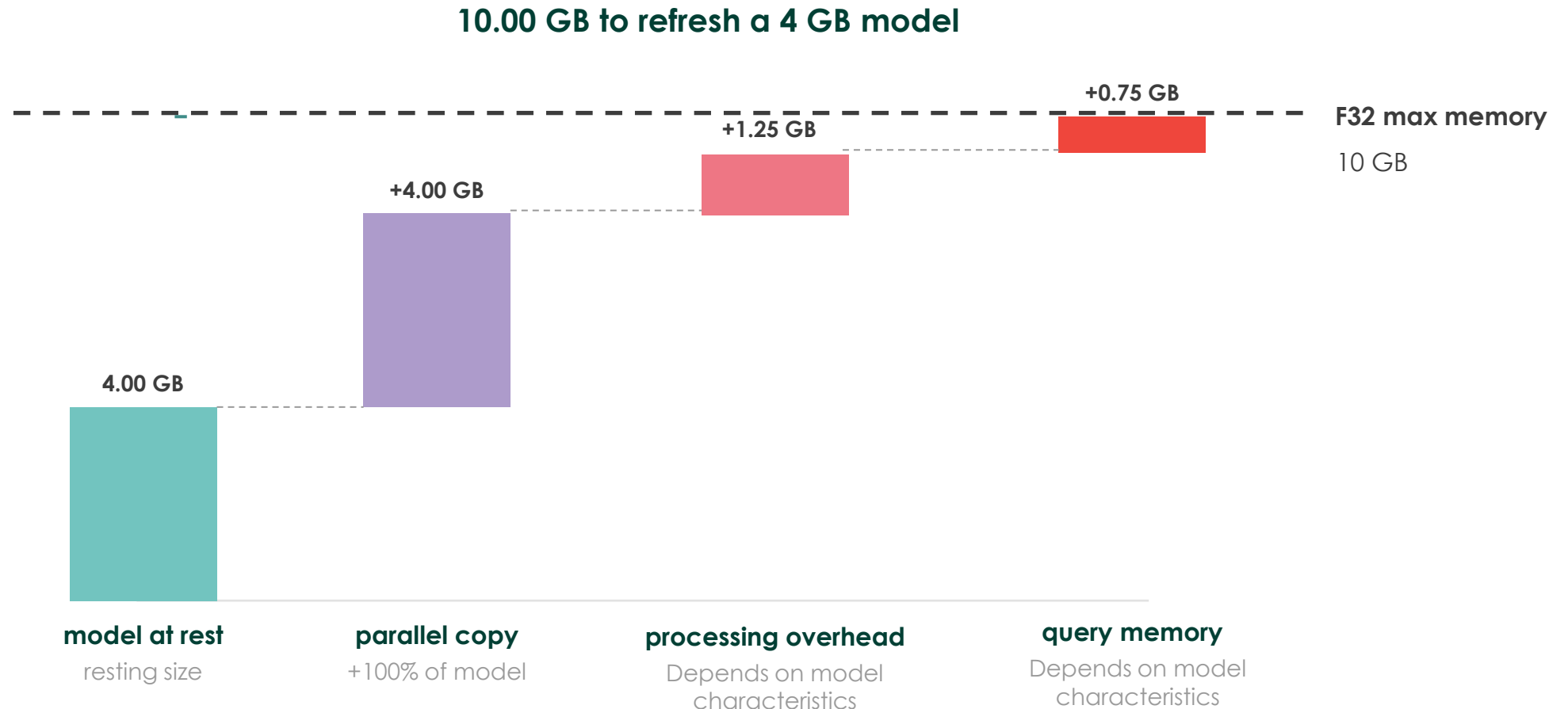
	F SKU	Power BI Pro	Power BI PPU
Fabric items	✓	✗	✗
Deployment pipelines	✓	✗	✓
DirectLake	✓	✗	✗
Memory limit	* Dependent on SKU	1 GB	100 GB
Cost	Price depends on SKU	€ 12,10 user/month	€ 20,80 user/month

Max memory per semantic model



Semantic model limitations

A full refresh needs about 2.5x the resting size, so plan for about 40% of the ceiling

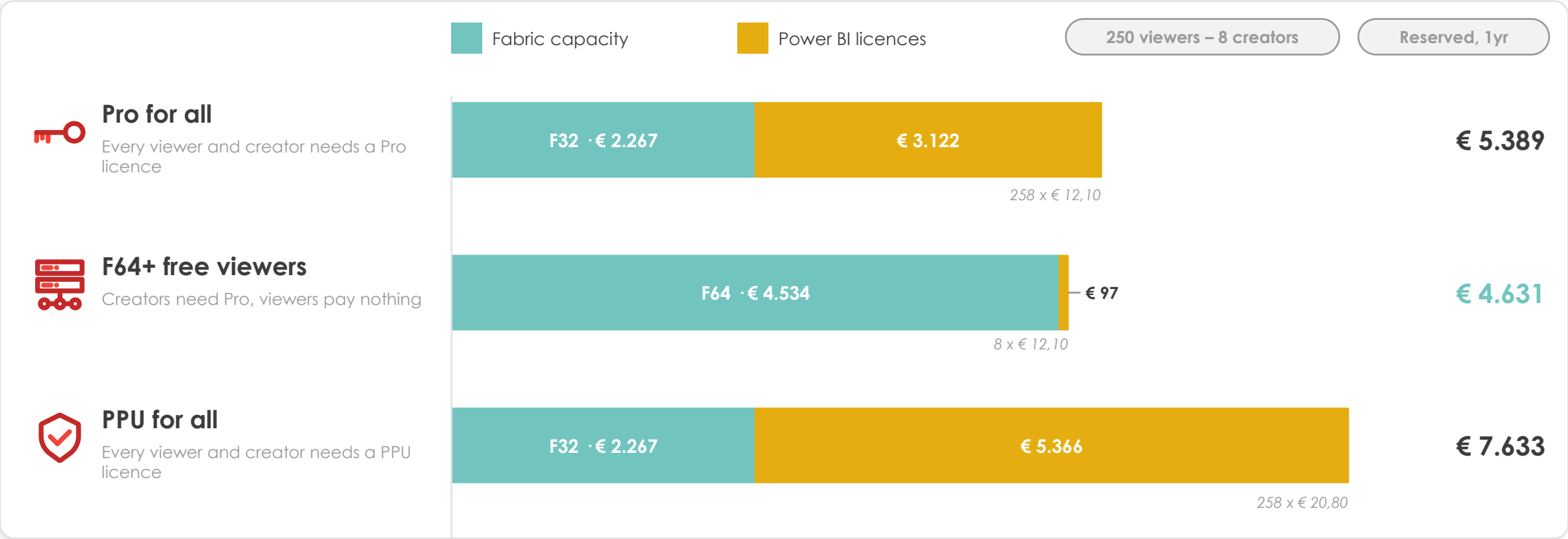


[Tabular Editor: optimizing semantic model memory in Fabric](#)



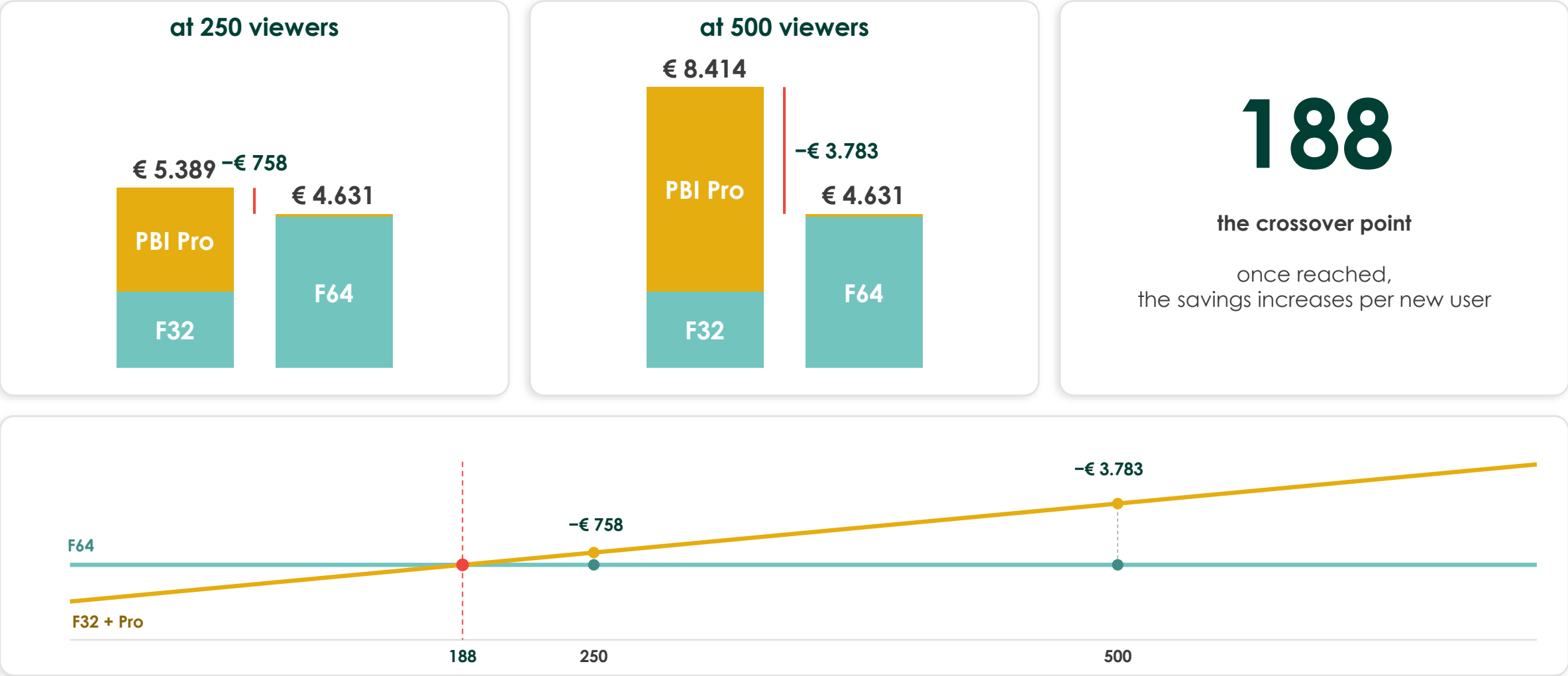
Licensing: three possible paths

Power BI Pro € 12,10
PPU € 20,80
per user per month



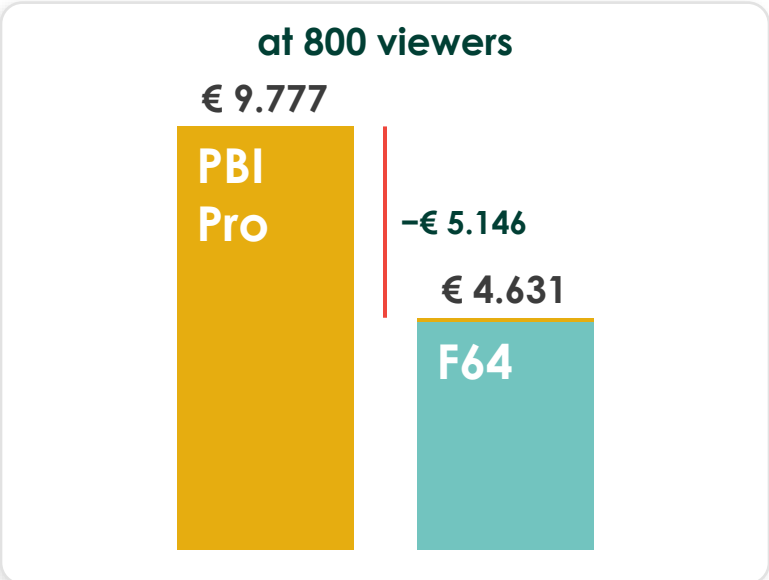
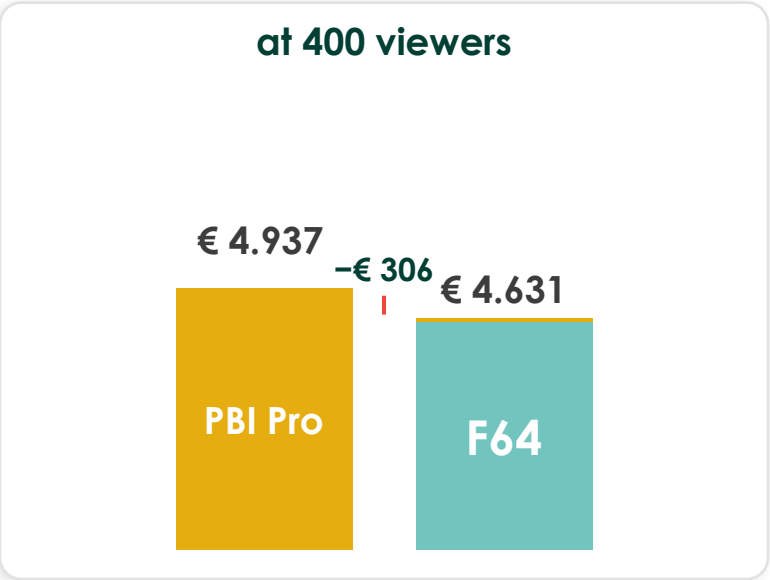
Crossover: the free-viewer threshold

The extra capacity pays itself in free viewers above this threshold



Crossover: the free-viewer threshold

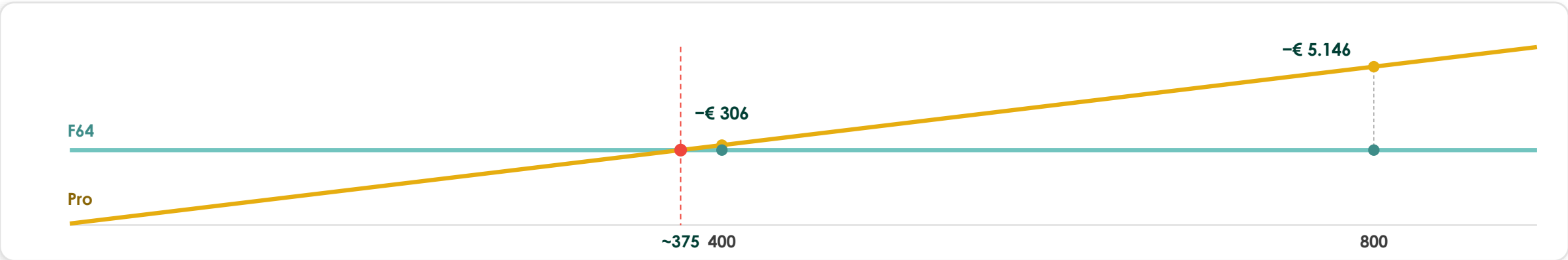
With Power BI only (no Fabric capacity needed) – higher threshold but still an option



375

the crossover point

once reached,
the savings increases per new user

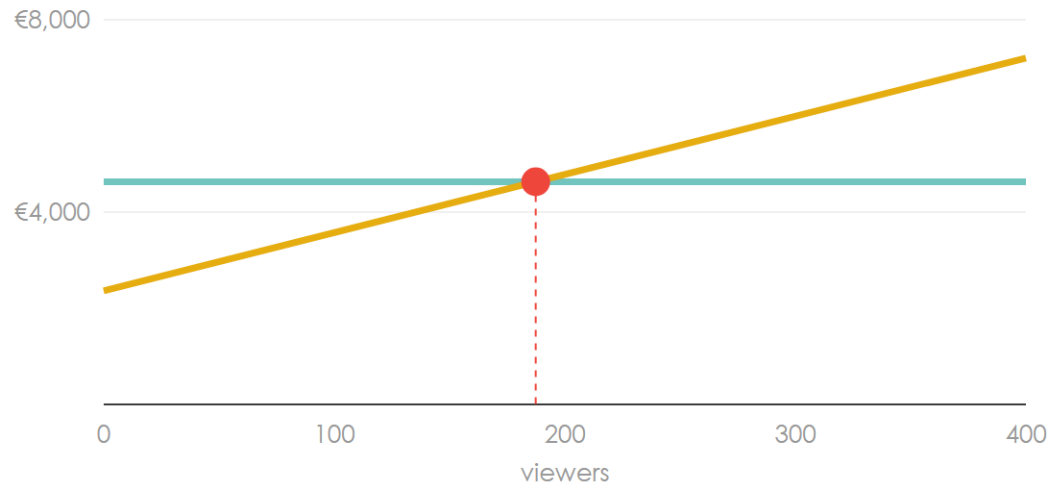


The threshold moves with the region

North Europe

F32 €2,267 · F64 €4,534 per month

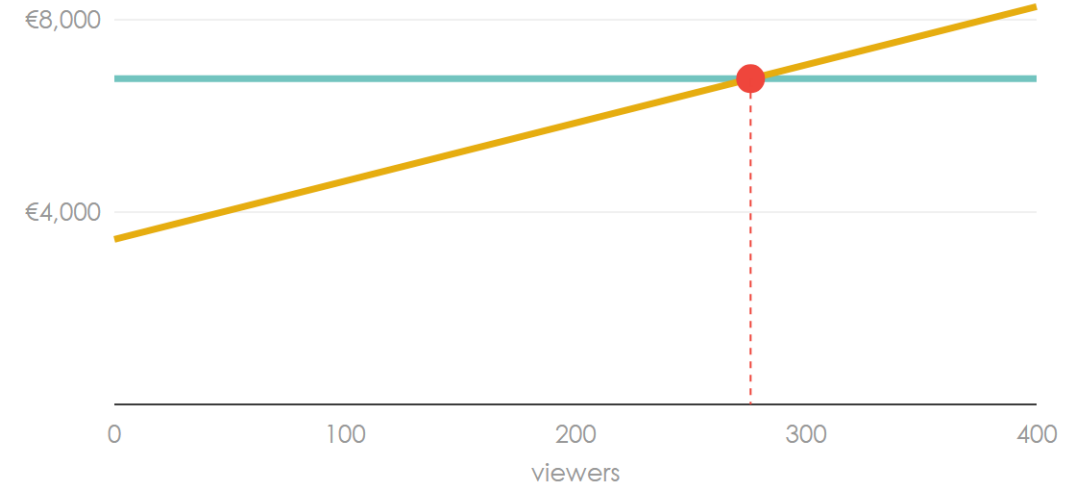
188 viewers



Brazil South

F32 €3,338 · F64 €6,677 per month

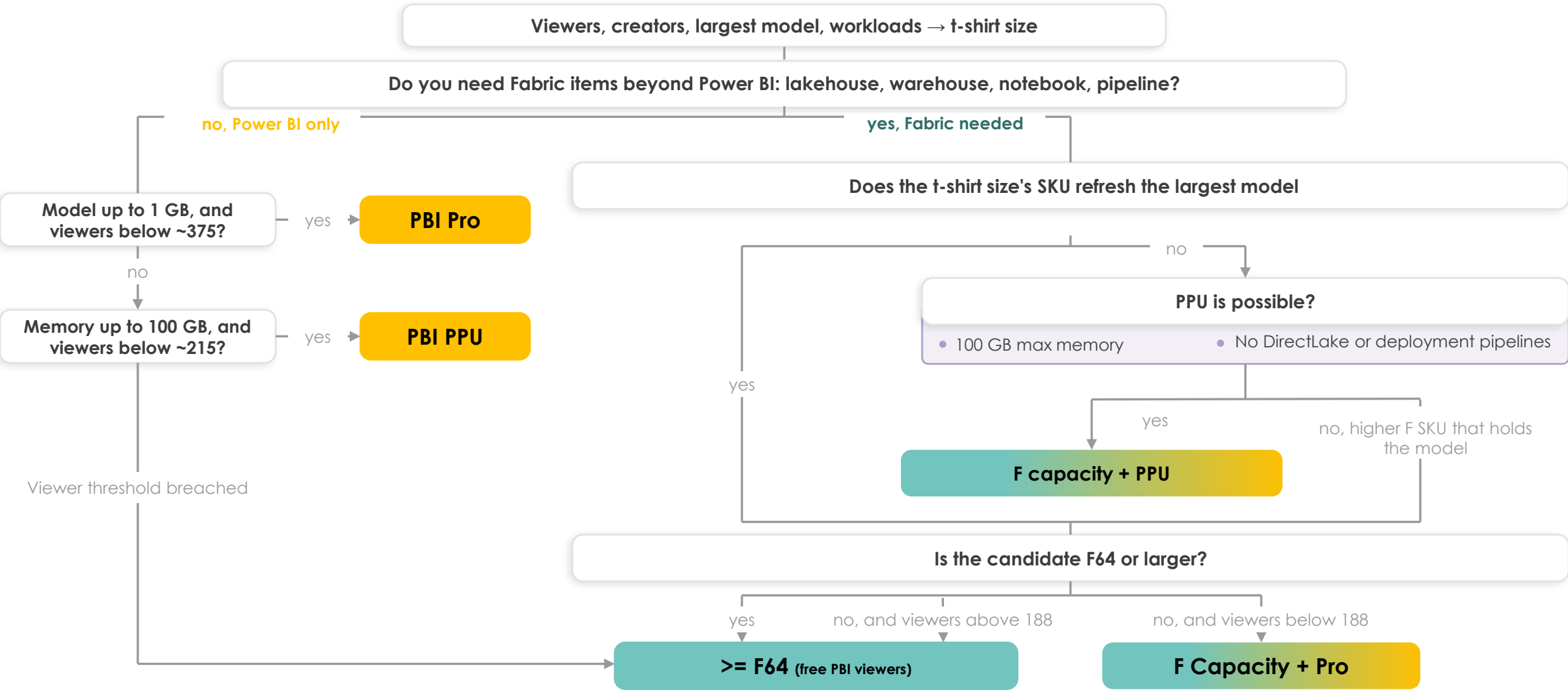
276 viewers



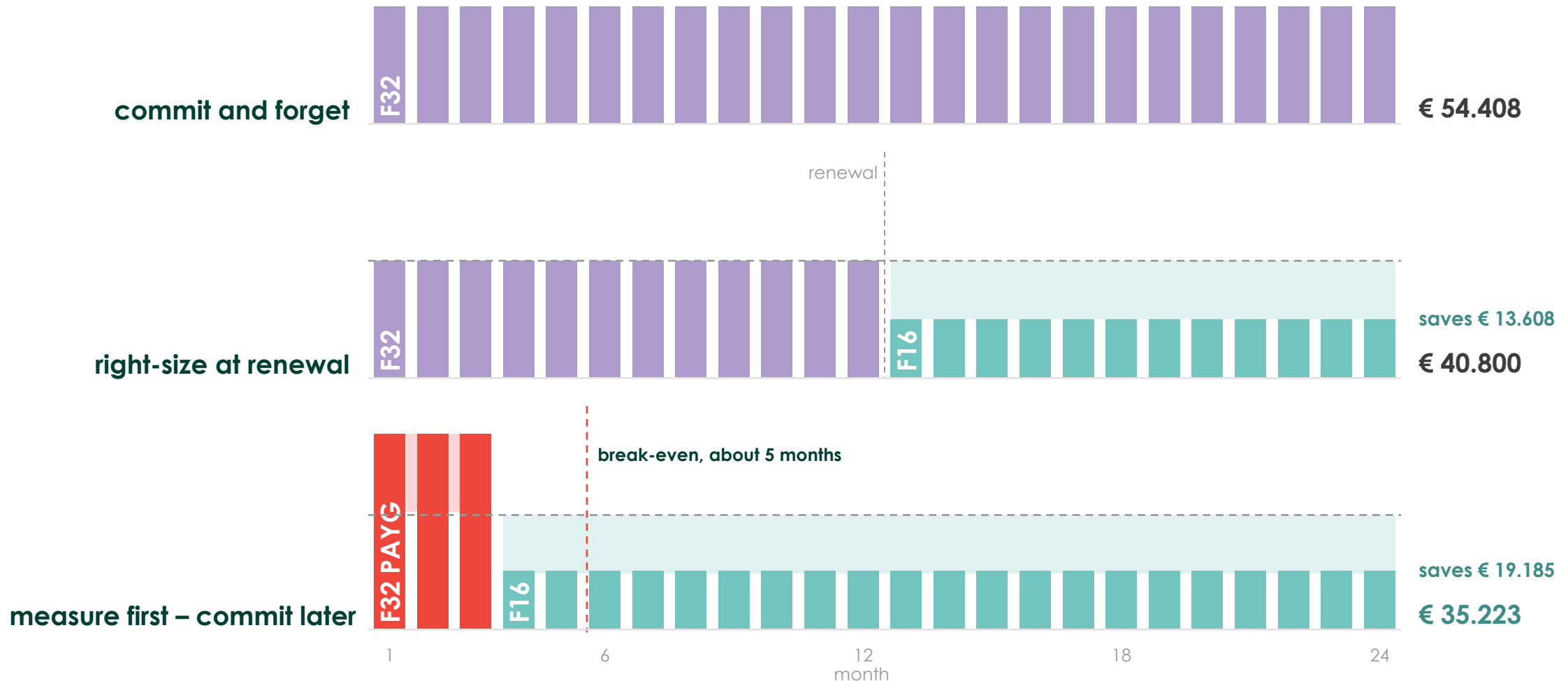
Note:

Capacity price follows the capacity region
Power BI licence price follows the tenant region

Licensing decision tree

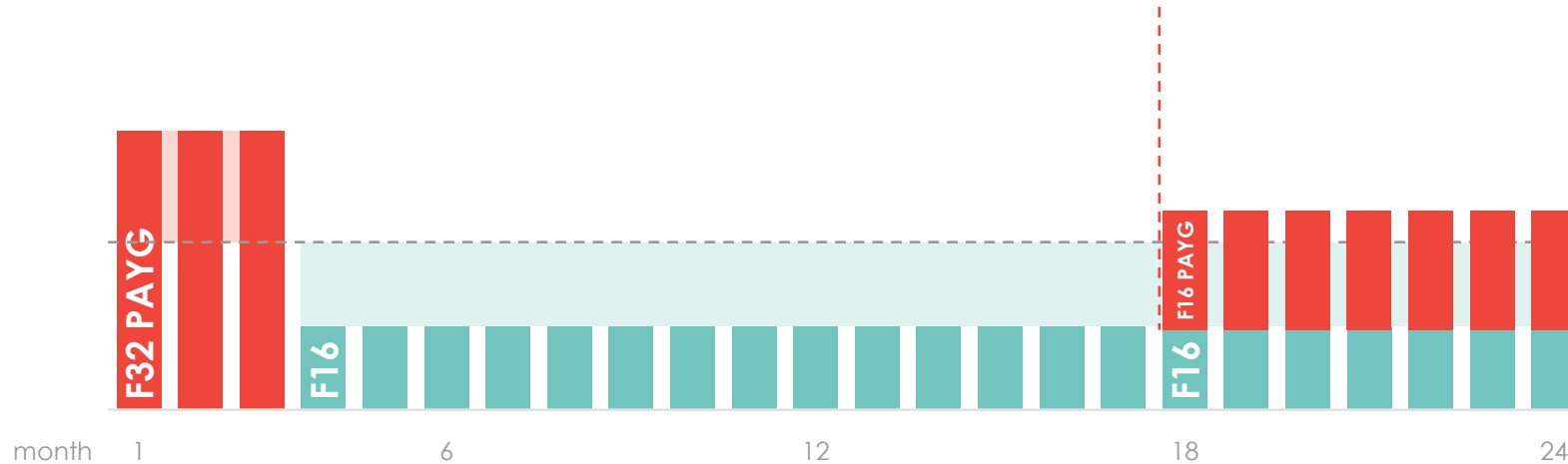


Start on pay-as-you-go, then commit



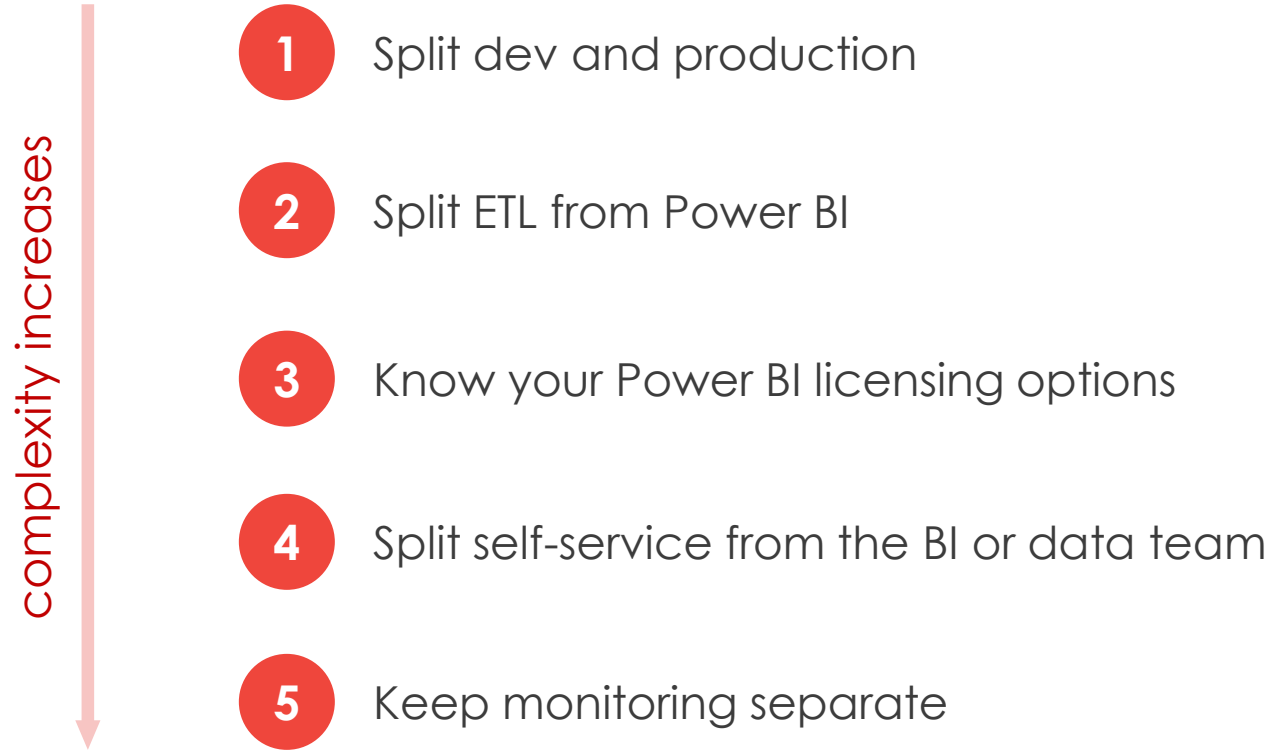
Start on pay-as-you-go, then commit

When a scale-up is needed before end of billing cycle, only surplus needs to be paid PAYG



[Reservation discounts](#)
[Reservation exchange process](#)

Capacity planning



Workspace tier

mission-critical

needs to stay on, always. For critical reports, RTI workloads, ...

operational

day-to-day reporting. Important to have high-availability but not critical if briefly interrupted

sandbox

assume this can be paused or noisy neighbours are present that influence each other

The workspace map

workspace

A workspace is linked to one capacity.

tier

What is the type of the capacity – is it mission critical or purely operational

env

Split dev from production. Dev/test/UAT are the cheapest workspaces to pause.

usage pattern (*)

see later: identify usage patterns

dominant workload

What drives the CU in this workspace

users

Important to know user volume – steers the licensing decision



capacity

We assign a capacity to every workspace

Map the workspaces, then assign capacities

workspace	tier	env	usage pattern	dominant workload	users	capacity
finance	mission-critical	prd	fully smoothed	semantic models	~120	
finance	operational	uat	consistently under-used	semantic models	~10	
sales	operational	prd	interactive bursts	semantic models	~85	
supplychain-etl	operational	prd	background ramp-up	pipelines	~15	
supplychain-etl	operational	dev	consistently under-used	pipelines	~8	
iot-realtime	mission-critical	prd	sustained overage	real-time	~30	
datascience-lab	sandbox	dev	under-used, spiky	ai / notebooks	~8	
monitoring	operational	prd	fully smoothed	semantic models	~5	

tier: ■ mission-critical ■ operational ■ sandbox

Result: capacity & workspace plan

Small tier

 **core-capacity**
F4 · reserved · prd / dev

 **supply-lakehouse-prd**
operational

 **supply-etl-prd**
operational

 **supply-etl-dev**
sandbox

 **Power BI shared capacity**
Microsoft-managed

 **finance-model-prd**
mission-critical Pro

 **finance-reports-prd**
mission-critical Pro

 **sales-model-prd**
operational Pro

 **sales-reports-prd**
operational Pro

Result: capacity & workspace plan

Medium Tier

 **core-capacity**
F64 · reserved · prd

 **finance-model-prd**
mission-critical

 **finance-reports-prd**
mission-critical

 **sales-model-prd**
operational

 **sales-reports-prd**
operational

 **supply-etl-prd**
operational

 **platform-monitoring**
operational

 **nonprd-capacity**
F8 · pay-as-you-go, paused weekends · uat / dev

 **finance-model-uat**
operational

 **finance-copilot-dev**
sandbox

 **supply-etl-dev**
sandbox

 **datascience-lab**
sandbox

 **Power BI shared capacity**

 **finance-reports-dev**
sandbox Pro

 **sales-reports-dev**
sandbox Pro

Result: capacity & workspace plan

Large Tier

 **critical-capacity**
F64 · reserved · prd

 finance-model-prd
mission-critical

 finance-reports-prd
mission-critical

 sales-model-prd
mission-critical

 sales-reports-prd
mission-critical

 **etl-capacity**
F32 · reserved · prd

 supply-lakehouse-prd
operational

 supply-etl-prd
operational

 marketing-dataflows-prd
operational

 platform-monitoring
operational

 **realtime-capacity**
F32 · reserved · prd

 iot-eventstream-prd
mission-critical

 iot-eventhouse-prd
mission-critical

 iot-model-prd
operational

 iot-reports-prd
operational

 **nonprd-capacity**
F32 · pay-as-you-go, paused weekends · uat / dev

 finance-model-uat
operational

 finance-reports-uat
operational

 supply-etl-dev
sandbox

 datascience-lab
sandbox

 **Premium Per User**

 finance-powerusers
operational

PPU

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Result: capacity & workspace plan

Xtra Large Tier

critical-capacity

F256 · reserved · prd

group-model-prd
mission-critical

group-reports-prd
mission-critical

opco-a-model-prd
mission-critical

opco-a-reports-prd
mission-critical

group-copilot-prd
operational

etl-capacity

F128 · reserved · prd

group-lakehouse-prd
operational

group-warehouse-prd
operational

supply-etl-prd
operational

marketing-dataflows-prd
operational

platform-monitoring
operational

realtime-capacity

F64 · reserved · prd

iot-eventstream-prd
mission-critical

iot-eventhouse-prd
mission-critical

iot-lakehouse-prd
operational

iot-model-prd
operational

iot-reports-prd
operational

nonprd-capacity

F64 · pay-as-you-go, paused weekends · uat / dev

group-model-uat
operational

group-reports-uat
operational

supply-etl-dev
sandbox

datascience-lab
sandbox

datascience-spark-dev
sandbox

Plan

1

t-shirt size

Choose a capacity based on t-shirt sizes

- user spread
- workload type
- platform size
- ...

2

license threshold

Check license needs based on users & estimated model size

- F64+ threshold
- PPU needs
- (number of) capacities needed

3

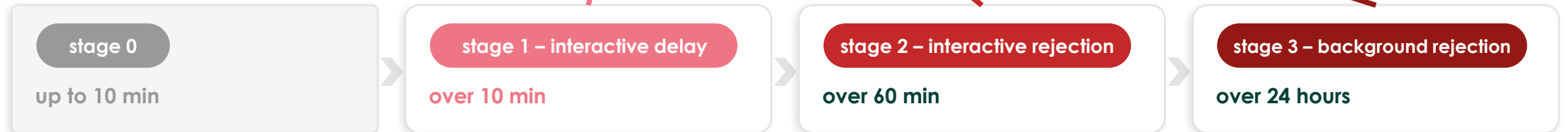
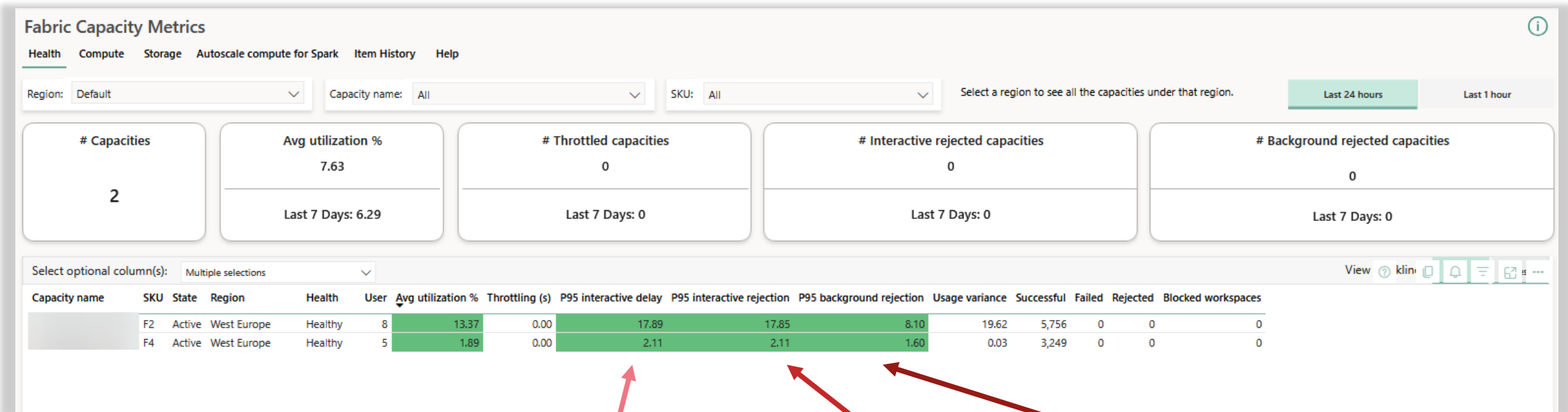
workspace mapping

Plan capacities based on workspace strategy

the five rules of thumb, applied on the workspace map

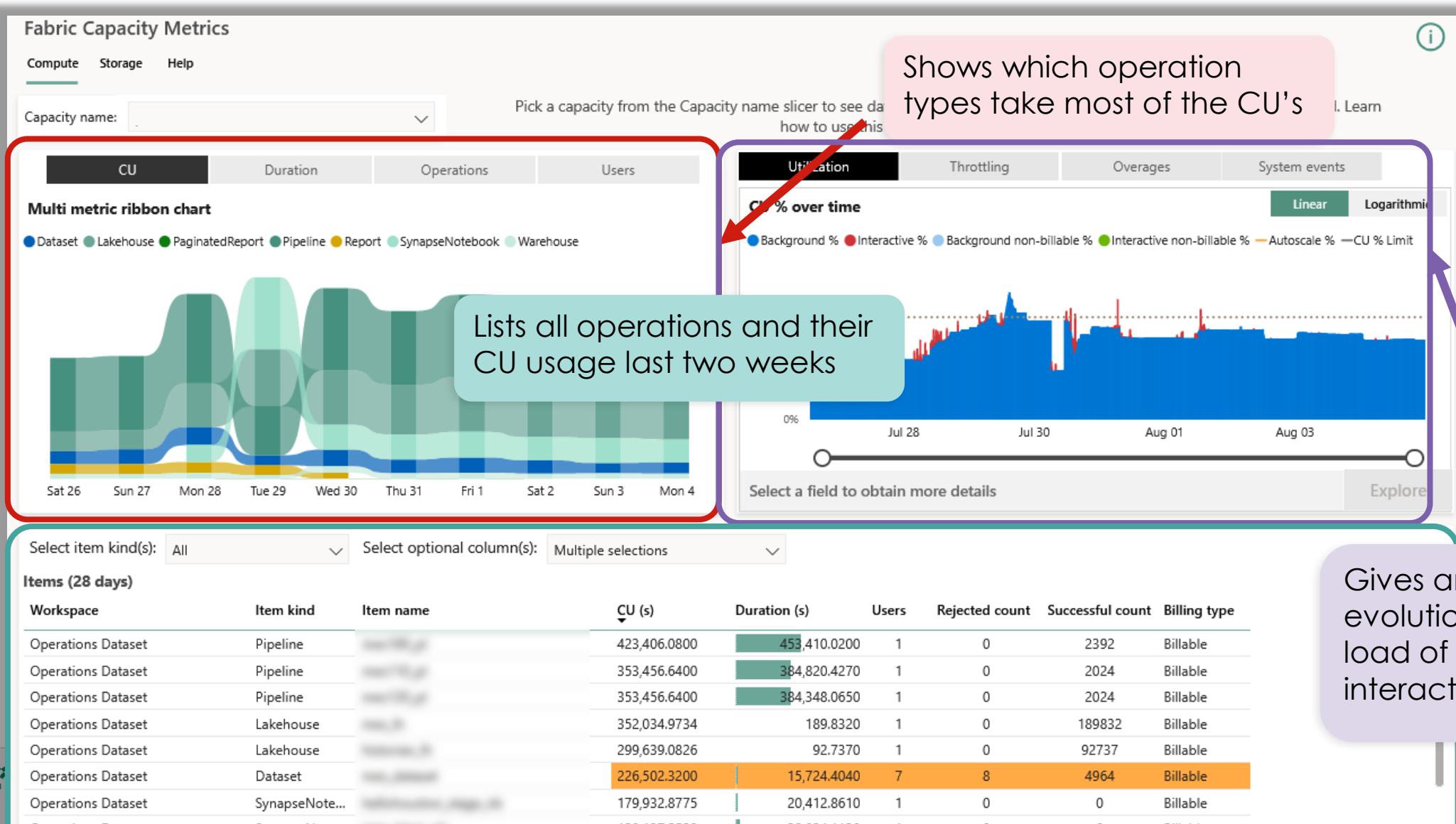
monitor: diagnose the capacity

The capacity metrics app



The capacity metrics app

Overview



Shows which operation types take most of the CU's

Lists all operations and their CU usage last two weeks

Gives an overview of evolution of the capacity load of background & interactive operations

The capacity metrics app

Back to report	ITEMS (14 DAYS)			
space	Item kind	Item name	CU (s)	Duration (s)
	Dataset		3,416,252.9504	71,512.9850
	EventStream		2,779,502.2454	4,644,000.0000
	Dataset		2,629,020.0000	53,083.1020
	Dataset		2,427,876.0736	110,982.3480
	KustoEventHouse		2,421,360.0000	1,076,160.0000
	Dataflow		2,227,092.6080	7,133.4500
	Dataset		2,048,595.0176	62,015.8850
	Dataflow		1,606,591.2320	53,646.0690
	Dataset		1,411,460.2016	33,534.1390
	Dataset		1,337,825.4048	50,669.1580
	Dataset		1,249,125.2576	23,145.3560
	Dataflow		1,222,415.9680	100,616.3570
	SynapseNotebook		1,131,452.7451	36,950.9360
	Dataflow		888,048.0320	20,545.1030
	Dataset		884,298.4192	56,896.5310
	Dataflow		856,329.0240	78,744.2380
	Dataset		717,641.5712	33,805.1950
	SynapseNotebook		653,107.4994	49,812.7630
	SynapseNotebook		566,256.6660	50,117.6230
	Dataset		544,425.5776	11,023.3580
	Dataset		517,685.0816	33,948.9230
	Dataflow		509,950.7040	41,721.2630
	Dataset		506,083.2288	16,693.4420
	Dataset		472,687.5808	7,853.0770
	Dataset		447,859.7344	14,789.2440
	Dataset		442,175.0336	12,406.9300
	Dataset		413,437.6320	16,039.9100
	Dataset		410,562.7808	16,359.6450
	Dataset		395,988.1280	10,077.0340

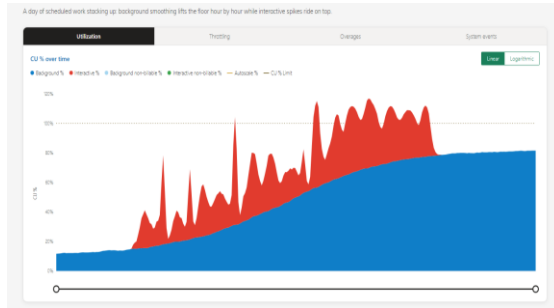
Item Performance

Compare CU's and see which operations have high usage or long duration

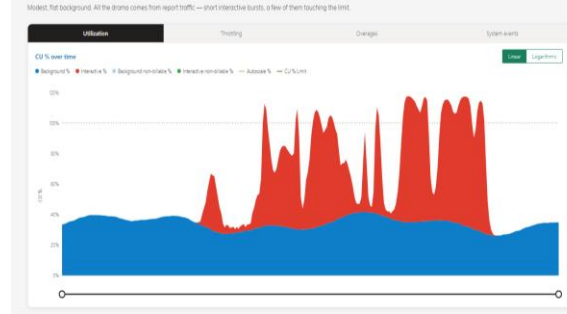
Overage patterns

24hr patterns

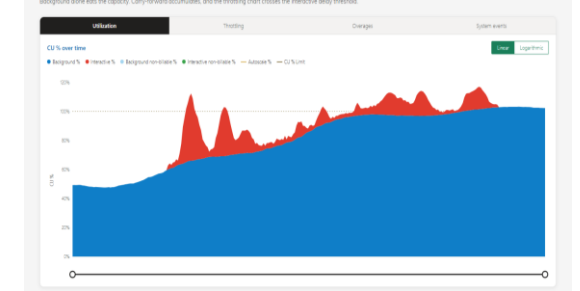
1 Background ramp-up



2 Interactive bursts

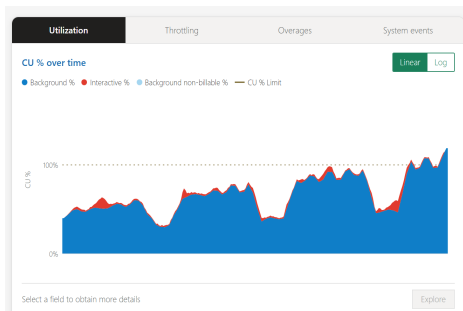


3 Sustained overage

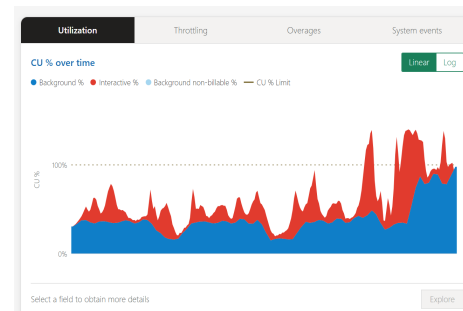


monthly patterns

4 Adoption creep



5 Month-end close



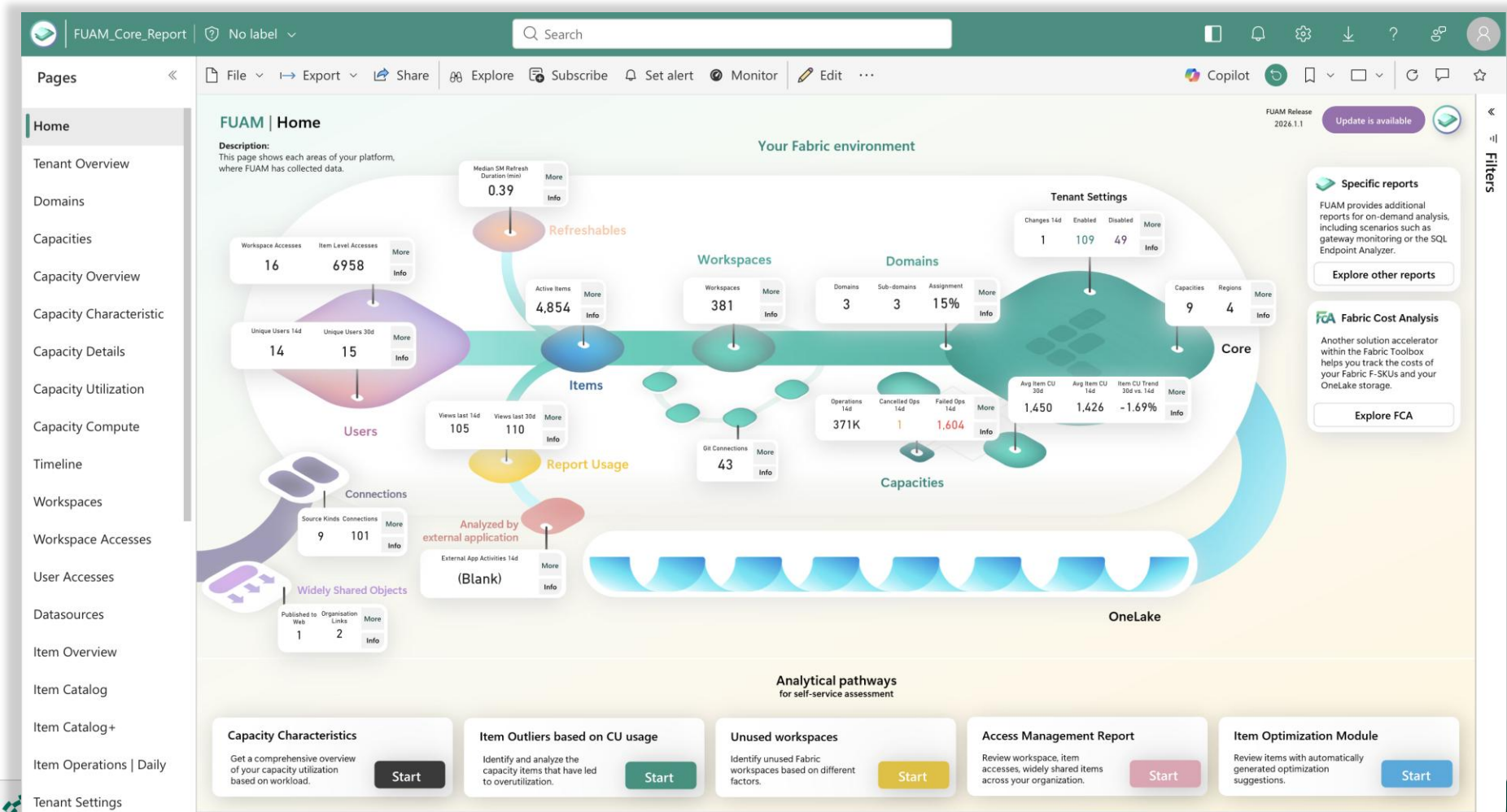
6 Consistently under-used



Other tools



Monitor the complete fabric estate:
Capacity usage, Item usage, Licensing, ...

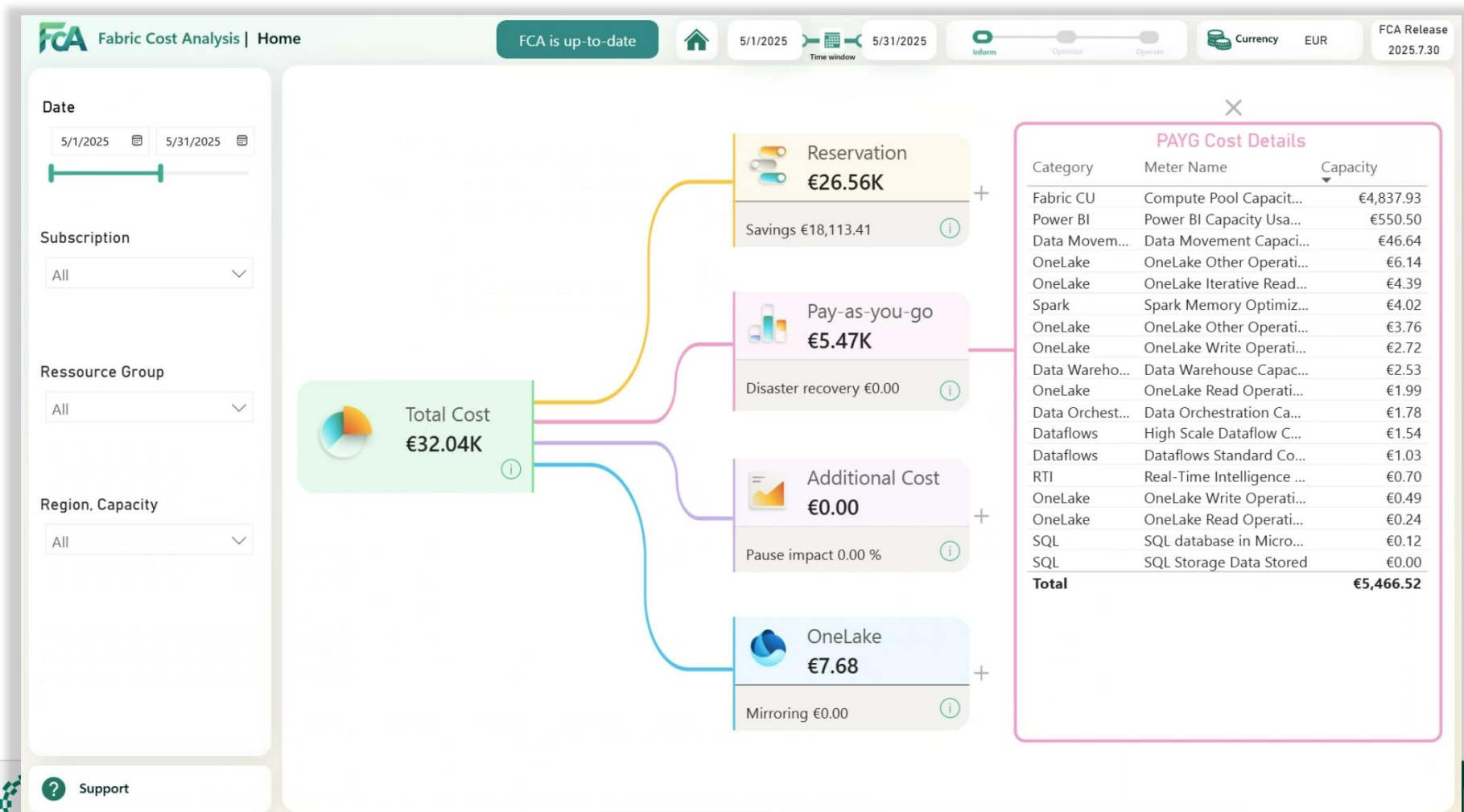


[fabric-toolbox > FUAM](#)

Other tools



Check storage costs / Chargeback
Assess reservation vs. pay-as-you-go



[fabric-toolbox > FCA](#)

Other tools



Capacity Events in real-time hub

React & alert on real-time events
Act before an overage occurs

events in real-time hub

- a **usage summary** every 30 seconds
- a **state change** the instant a capacity goes overloaded, is paused, or resumes.



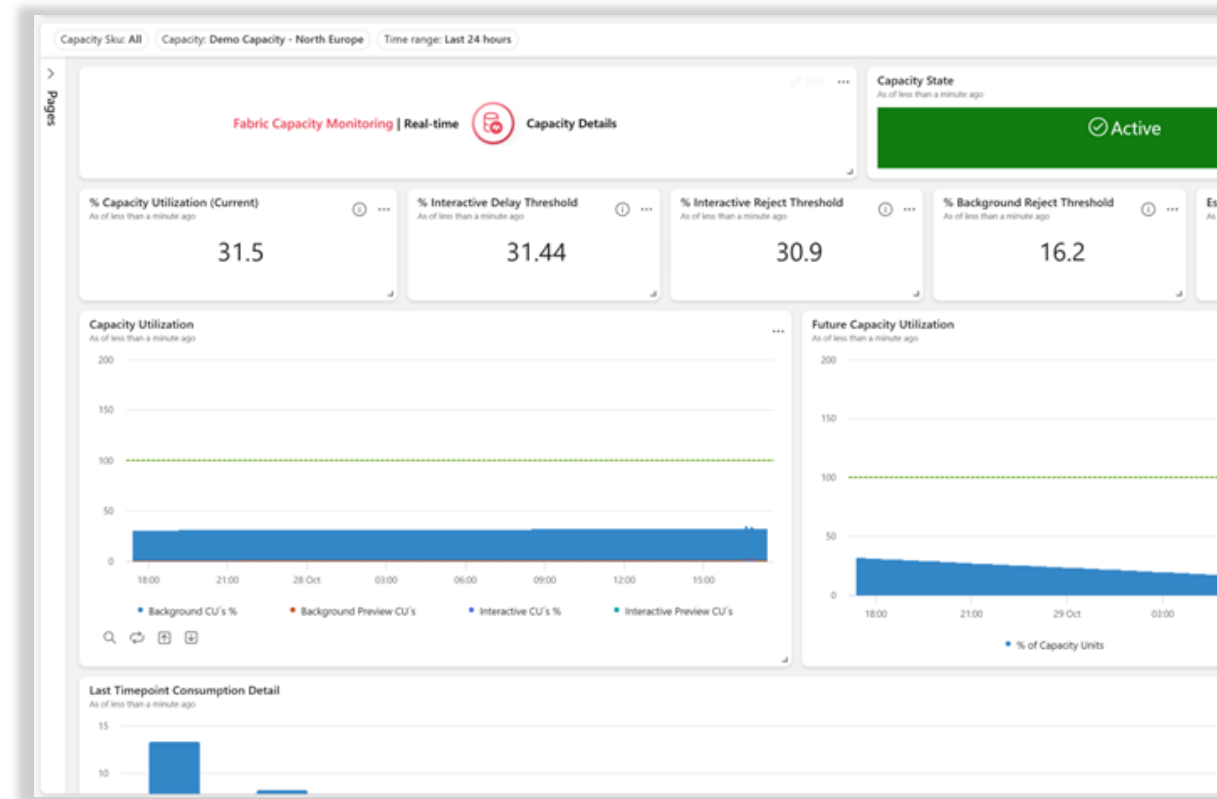
activator

Turns an event into a rule



eventstream + eventhouse (optional)

Route events to a KQL database for history and analytics



Other tools



Azure Cost Monitor

Check azure costs & invoices

Home > Cost Management + Billing | Billing scopes > Cost Management + Billing | Invoices

Cost Management + Billing | Invoices

Microsoft

Search (Ctrl+ /)

Overview

Cost Management

Diagnose and solve problems

Billing

Subscriptions

Invoices

Properties

Payment methods

Support + troubleshooting

New support request

Email Invoice

Access Invoice

How satisfied are you with invoices? →

Here are the invoices for subscriptions of which you are the Account Administrator. To view invoices for subscriptions for which you have [role-based access control \(RBAC\) permissions](#) to manage Azure resources, go to [Subscriptions](#). To view usage for the current billing period in Cost analysis [click here](#).

Subscri... : Visual Studio Enterprise

Timespan : Last 3 months

Billing Period	Invoice date	Amount	Type	Download
1/26/2020-2/25/2020	2/26/2020	USD 0.00	Azure Services	↓
12/26/2019-1/25/2020	1/26/2020	USD 0.00	Azure Services	↓
11/26/2019-12/25/2019	12/26/2019	USD 0.00	Azure Services	↓

Source: docs.azure.com

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Other tools



M365 Admin center

Monitor Power BI Licenses

Microsoft 365 admin center

Search

Dark mode

All tenants

Home

Users

Devices

Teams & groups

Billing

Purchase services

Your products

Licenses

Bills & payments

Billing accounts

Payment methods

Billing notifications

Setup

Show all

Stratada

Licenses

SubscriptionsRequestsAuto-claim policy

Select a product to view and assign licenses. Each product below may contain licenses from multiple subscriptions. [Learn more about assigning licenses](#)

Go to Your products to manage billing or buy more licenses.

ExportRefreshOrganization

Name ↑	Available licenses	Assigned licenses	Account type
Azure Active Directory Basic	17	8/25	Organization
Dynamics 365 Customer Engagement Plan	7	8/15	Organization
Microsoft 365 Audio Conferencing	0	4/4	Organization
Microsoft 365 E3	6	19/25	Organization
Microsoft Power Apps Plan 2 Trial	9992	8/10000	Organization

Source: docs.azure.com

Monitor recap

Use a combination of tools to monitor

 Fabric capacity metrics app

 FUAM

 Real-time hub

 Fabric Cost Analysis

 Azure cost monitor

 M365 admin center

Identify overage patterns

- Identify symptoms and root causes
- List offending workloads & workspaces as optimisation targets

optimise

Optimise: mitigate, scale, or isolate



Mitigate first

Fix the concurrency or workload problem



Scale

Increase capacity or move to a bigger SKU



Isolate

Move the workload onto its own capacity

Optimise costs

Optimise step 1 - Mitigate

Mitigate: low hanging fruit

Look for the big CU hogs

List high-consuming items – do they use more CU than expected? This can be due to:

- Workload type
- Frequency
- Volumes
- ...

Typical offenders and their solutions

Semantic models

- Use Incremental load
- Simplify DAX
- Avoid too much transformations in model
- Reduce Frequency

Dataflows Gen 2

- Apply query folding
- Change to alternative engine:
 - PySpark/Python
 - ADF & Stored Procedures
- Reduce Frequency

Spark

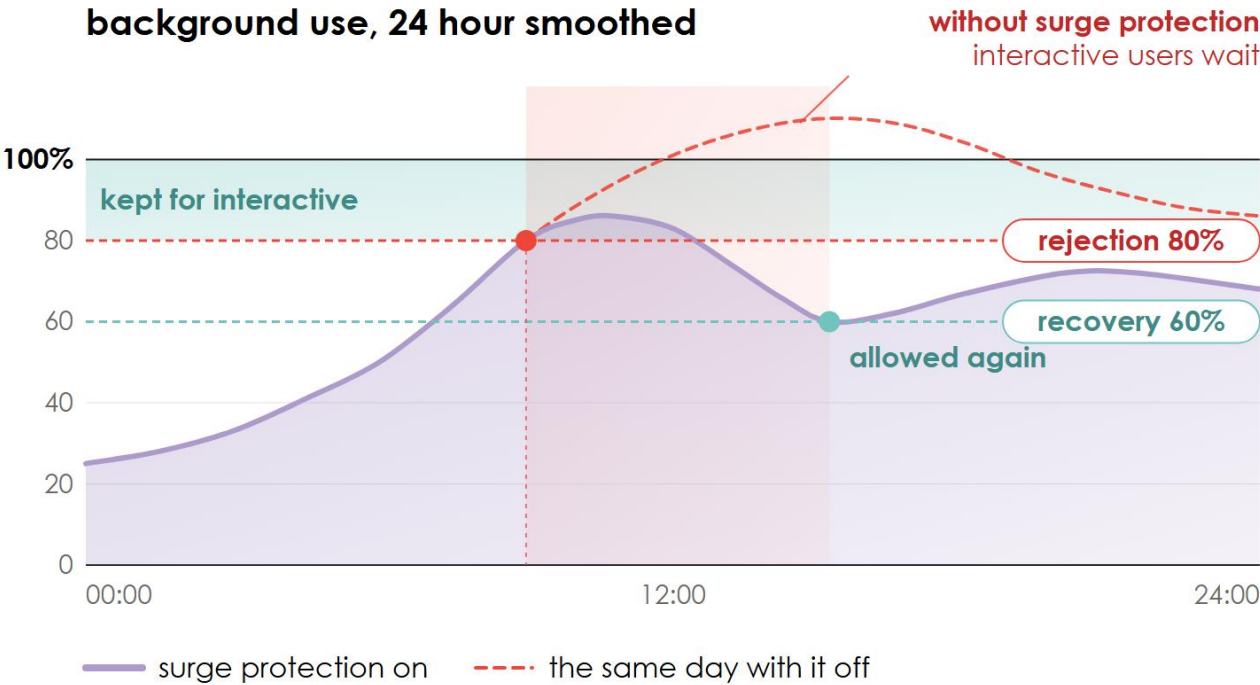
- Right-size your Spark pools
- Utilize concurrent sessions

...

< Back to report | ITEMS (14 DAYS)

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	Dataset		3,416,252.9504	71,512.9850	
	EventStream		2,779,502.2454	4,644,000.0000	
	Dataset		2,629,020.0000	53,083.1020	
	Dataset		2,427,876.0736	110,982.3480	
	KustoEventHouse		2,421,360.0000	1,076,160.0000	
	Dataflow		2,227,092.6080	7,133.4500	
	Dataset		2,048,595.0176	62,015.8850	
	Dataflow		1,606,591.2320	53,646.0690	
	Dataset		1,411,460.2016	33,534.1390	
	Dataset		1,337,825.4048	50,669.1580	
	Dataset		1,249,125.2576	23,145.3560	
	Dataflow		1,222,415.9680	100,616.3570	
	SynapseNotebook		1,131,452.7451	36,950.9360	
	Dataflow		888,048.0320	20,545.1030	
	Dataset		884,298.4192	56,896.5310	
	Dataflow		856,329.0240	78,744.2380	
	Dataset		717,641.5712	33,805.1950	

Mitigate: surge protection



Capacity-level

Protects interactive workloads from background workloads competing for the same CU.

Surge Protection

Unapplied changes

Workspace Consumption (Preview)

Set a maximum percentage of capacity that any single workspace can consume for background operations. Lower limits help prevent any one workspace from monopolizing capacity in a rolling 24-hour window. High-priority capacities can be excluded. [Learn more](#)

☐ Off

Background Operations

When total CU consumption reaches the rejection threshold, reject new background operations. When CU consumption drops below the recovery threshold, accept new background operations.

☒ On

Rejection threshold

80 %

Recovery threshold

60 %

Apply

Discard

Mitigate: surge protection



Workspace-level (preview)

Caps individual workspaces' CU percentage.
No per-workspace control
Exceptions can be made for mission-critical capacities.

Surge Protection

Unapplied changes

Workspace Consumption (Preview)

Set a maximum percentage of capacity that any single workspace can consume across both interactive and background operations. Lower limits help prevent any one workspace from overusing capacity. Limits apply over a rolling 24-hour window. High-priority capacities can be excluded. [Learn more](#)



On

Rejection threshold

25 %

Block

For



12

Hours

Workspaces assigned to this capacity

Search for, add, or remove workspaces assigned to this capacity

Search workspaces

Remove all

Workspace name ↑

Workspace admins

Actions ⓘ

Status ↑↓

OneLake Geo-
replication

Monitoring

Availability



Data Platform PRD

[View admins](#)



Assigned

--

Inactive

Mission critical

Workspace settings

Data Platform PRD

Workspace availability

Control when this workspace can accept operation requests. [Learn more](#)



Available



Mission critical



Blocked

Optimise step 2 - Scale

Scaling: the 59% crossover



Below 59% of time

Pause + PAYG wins.



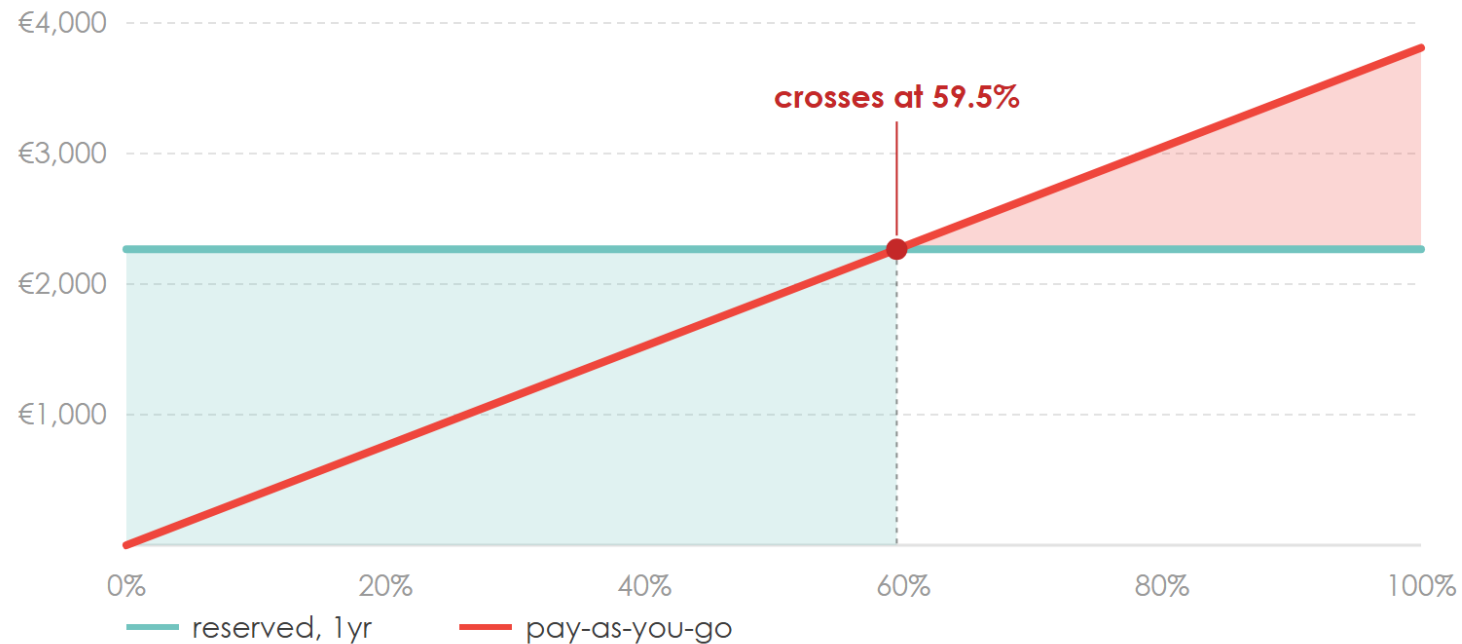
59%

breakeven



Above 59% of time

Reserve wins.



Scaling as routine

Two routes to automation

🕒 timed

Azure Automation schedule



PowerShell runbook



Update-AzFabricCapacity



the capacity scales

⚡ event-based

capacity event, every 30 seconds



Activator rule on the threshold



Fabric user data function



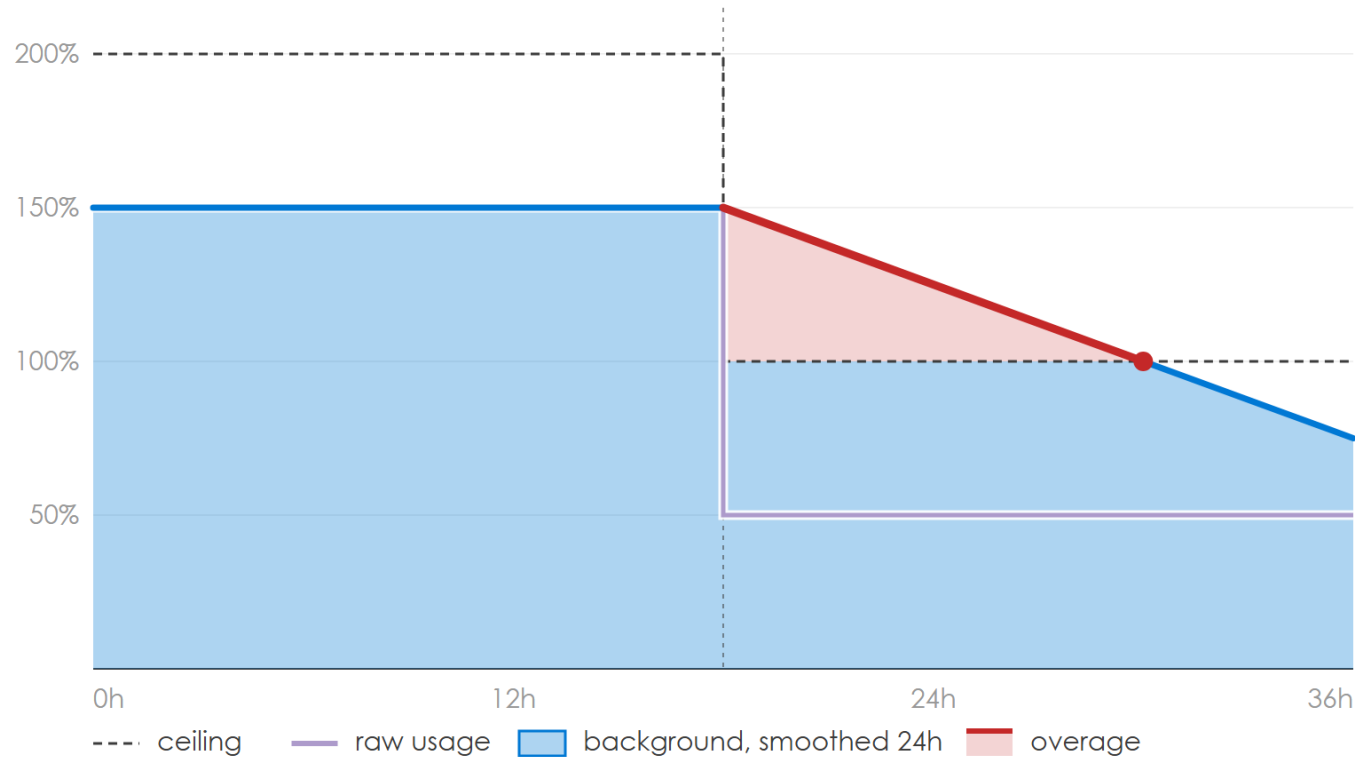
the capacity scales

Remarks

1. There is no such thing as autoscale
2. Scaling down comes with a risk (F64 & F512)

Scaling as routine

The background danger



only scale in whole days

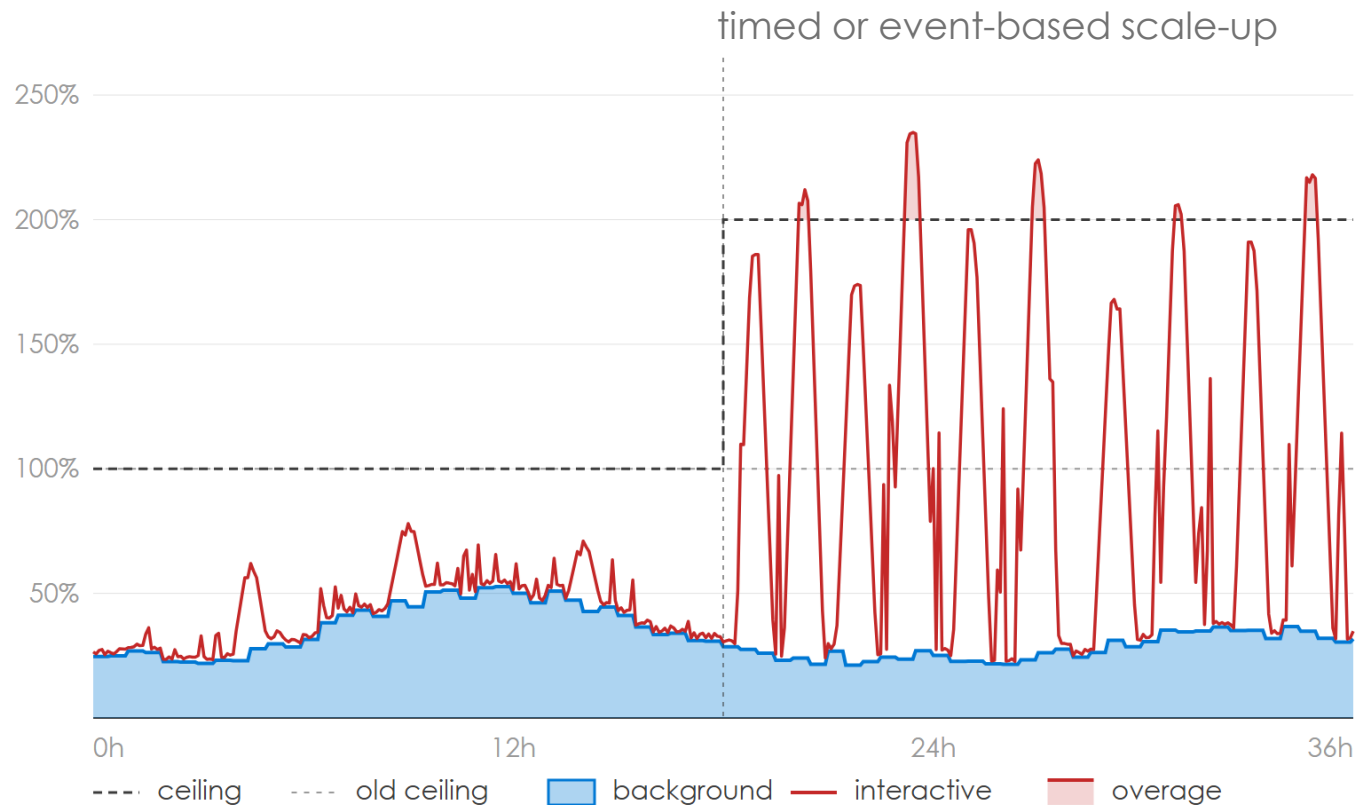
The safe version of intraday scaling to cater for background activities: operates in whole-day blocks, never hours.

alternative: pausing clears the debt immediately

at PAYG pricing
Running jobs are cancelled.

Scaling as routine

The interactive fix

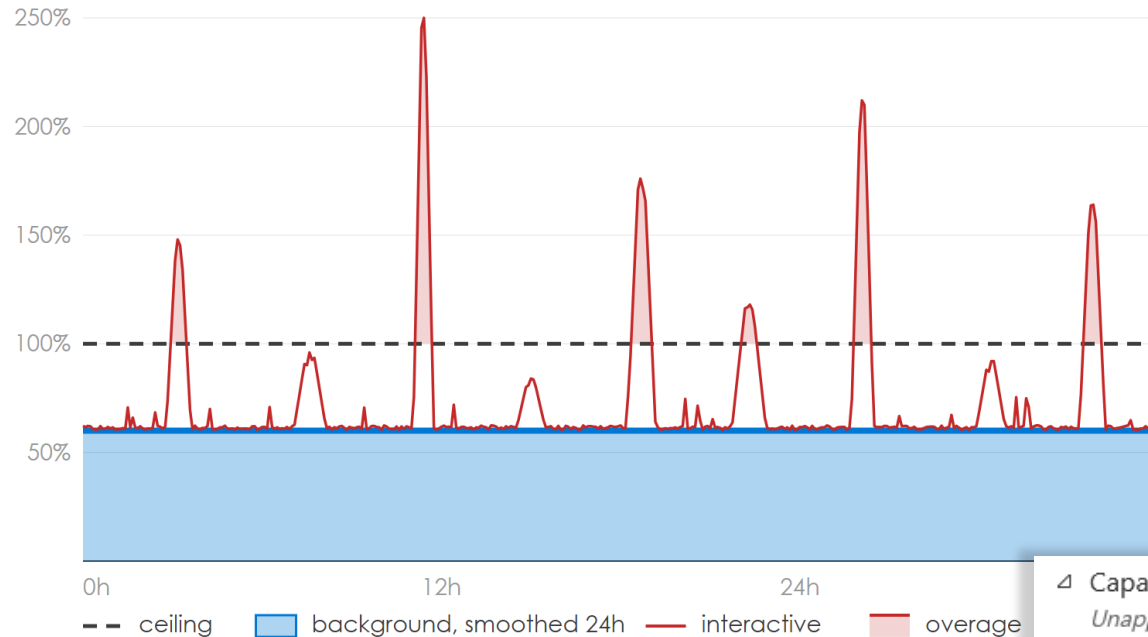


Scaling can happen immediately

Extra headroom from a scale-up is available almost immediately.
No effect on background activities – but does have effect for interactive

scaling as routine

Capacity overage (preview)



You only pay the excess

3x PAYG pricing

Ideal for spiky workloads or busy interactive periods

Safest option to ensure business continuity

Capacity Overage (Preview)

Unapplied changes

Fabric includes built-in Capacity Overage protection to help workloads run smoothly. When usage exceeds SKU limits, Capacity Overage provides additional Capacity Units (CUs) to avoid throttling, up to a 24-hour rolling threshold that can be adjusted at any time. All excess usage accrued before entering overage, along with operations that run during overage and for up to five minutes after the threshold is reached, avoid throttling and are billed at the overage rate. We recommend keeping Capacity Overage enabled to reduce the risk of throttling. [Learn more](#)



On

Rolling 24-hour overage limit



288



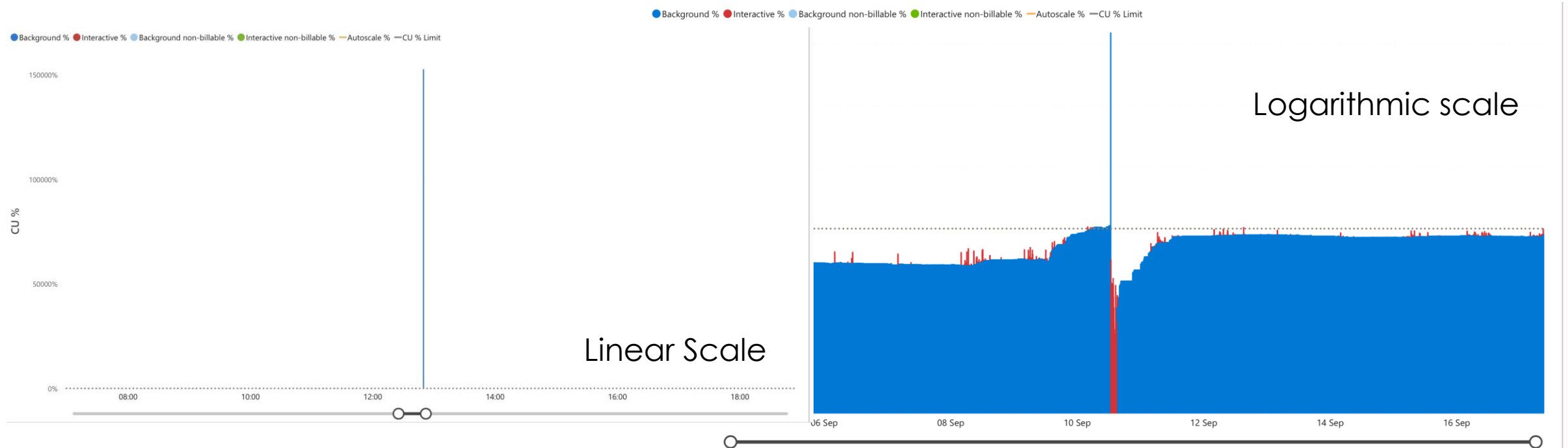
CUhr

Bill shock: scaling down vs. pause

Immediately pay for the “overage” and start with a clean capacity

Running jobs are cancelled

Causes “bill shock” → all future smoothed usage is paid back PAYG



Optimise step 3 - Isolate

Copilot capacity

Route your copilot usage to a separate capacity

- Enable Fabric Copilot capacities in tenant settings
- Assign the capacity to specific users or groups



Copilot and Azure OpenAI Service

- Users can use **Copilot** and other features powered by Azure OpenAI
Enabled for the entire organization
- Data sent to Azure OpenAI can be processed outside your capacity's geographic region, compliance boundary, or national cloud instance
Enabled for the entire organization

- ▾ Capacities can be designated as Fabric **Copilot** capacities
Enabled for the entire organization
With this setting on, capacity admins can designate capacities as Fabric **Copilot** capacities. **Copilot** capacities are special capacity types that allow your organization to consolidate users' **Copilot** usage and billing on a single capacity. [Learn More](#)
When users use **Copilot** features, capacity admins can see the names of the items associated with users' **Copilot** activity. [Learn More](#)

☒ Enabled

Apply to:

- ☒ The entire organization
- ☐ Specific security groups
- ☐ Except specific security groups

Apply

Cancel

▾ Copilot capacity

Enabled for a subset of the organization

Turn on this setting to designate this capacity as a Fabric Copilot capacity. Copilot capacities are special capacity types that allow your organization to consolidate users' Copilot usage and billing on a single capacity. Copilot capacities may not be available in all regions. [Learn more](#)

Select the users or user groups who can use this capacity for their Copilot usage and billing.

Apply to:

- ☐ The entire organization
- ☒ Specific users or groups

Clear all

AT Armando Test ✕ Fabrice Admin ✕

Apply

Discard

Autoscale billing for spark

(Autoscale) Billing for Spark Jobs

- Spark jobs use pay-as-you-go consumption outside your capacity
- Ideal to keep lower capacities performant (higher limits, separate billing)
- Setting on capacity level (capacity admin)

Autoscale Billing for Apache Spark

Unapplied changes

Turn on this setting to use a pay-as-you-go model for Apache Spark jobs. With autoscale turned on, you can define a maximum CU limit on Spark consumption. Bursting and smoothing aren't applied when autoscale is in effect.

Changes to this setting, including turning it on or off and reducing the maximum CU limit for Spark consumption, cancels all currently running jobs. [Learn more](#) 

☒ On

Maximum Spark CU consumption (1 CU = 2 Spark v-cores)

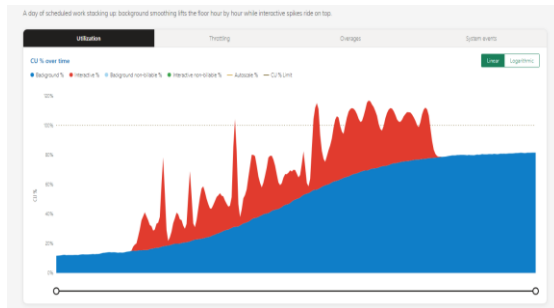
 24  Capacity Units

This setting uses 25% of your subscription's CU quota (32 of 128).
Need more? [Learn how to increase your quota.](#)

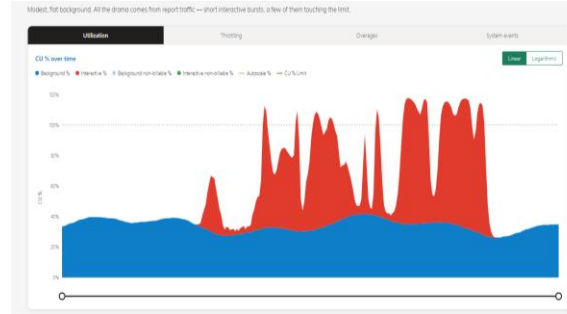
Optimise the different patterns

24hr patterns

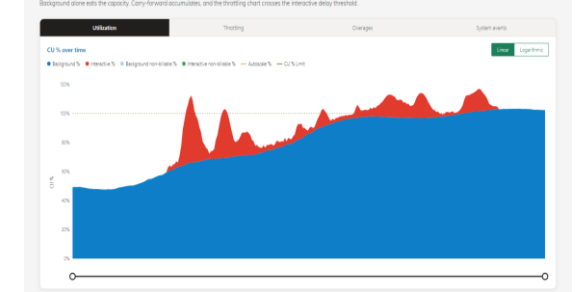
1 Background ramp-up



2 Interactive bursts

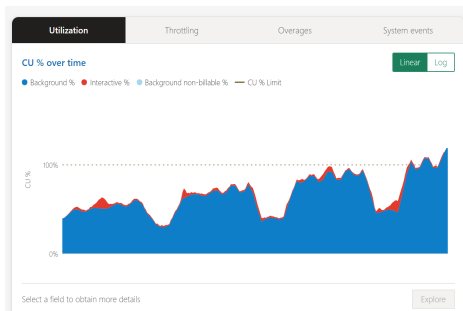


3 Sustained overage

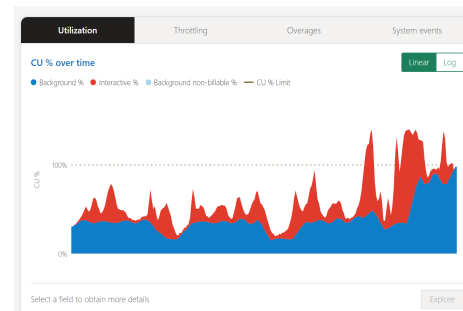


monthly patterns

4 Adoption creep



5 Month-end close

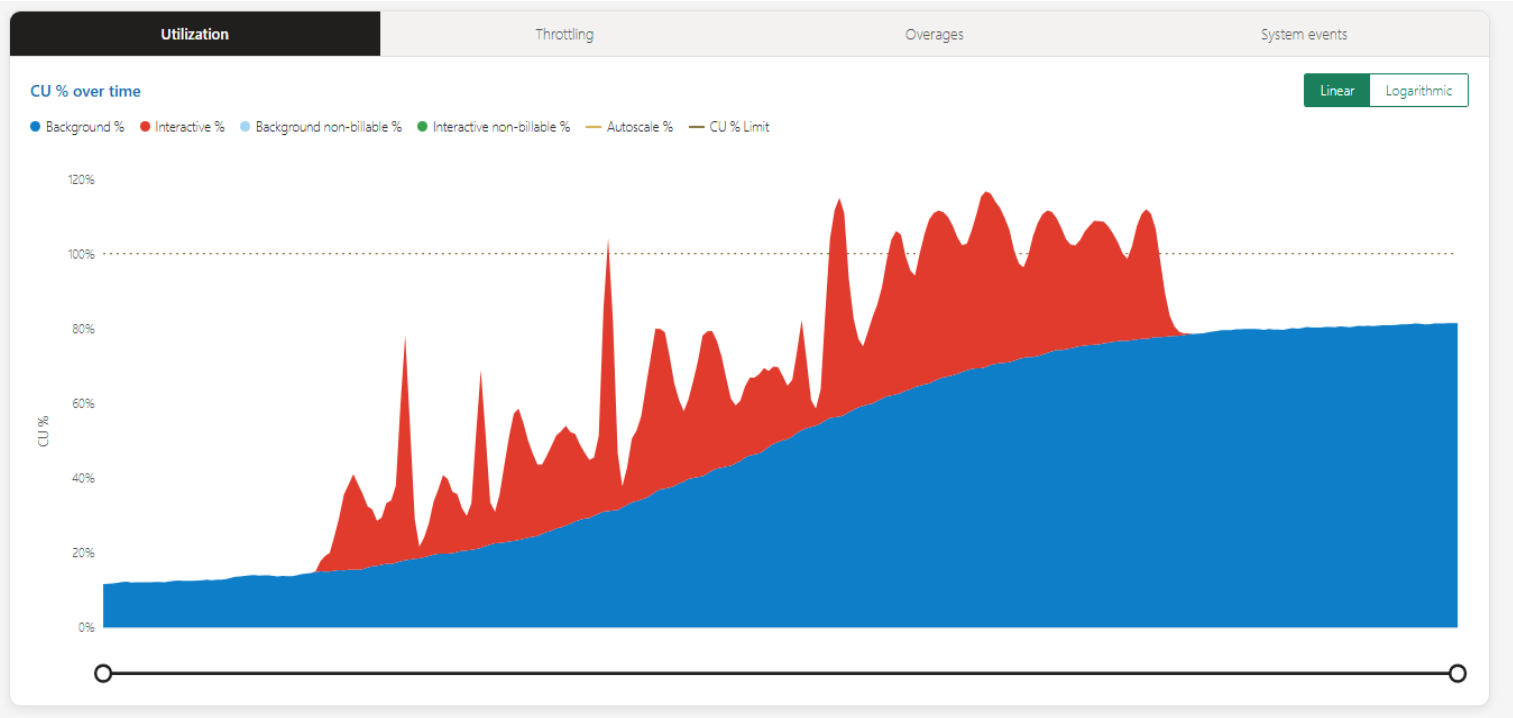


6 Consistently under-used



Background ramp-up

24hr pattern



Symptoms

Background usage climbs hour by hour through the day instead of holding flat, with interactive spikes on top. Only manageable if there's time to free up smoothed usage 24h after.

Possible reasons

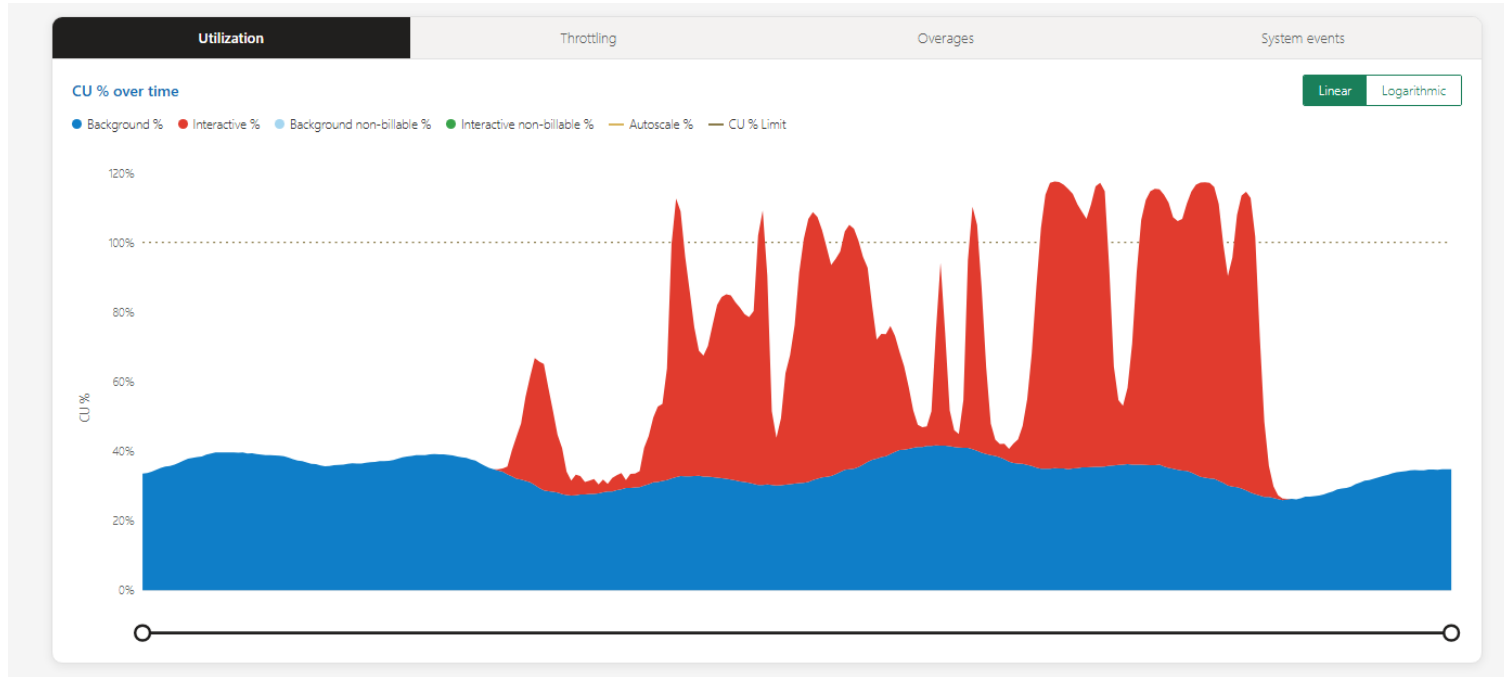
Background smooths over a rolling 24 hours. Jobs stack throughout the day. If new pipelines or exceptional refreshes are added on top, it gets stacked.

Mitigation

- If only sporadic throttling happens: capacity overage
- Otherwise: scale
- Move/change expensive workloads

Interactive bursts

24hr pattern



Symptoms

Background stays flat = ok
Interactive report usage breaks the ceiling.

Possible reasons

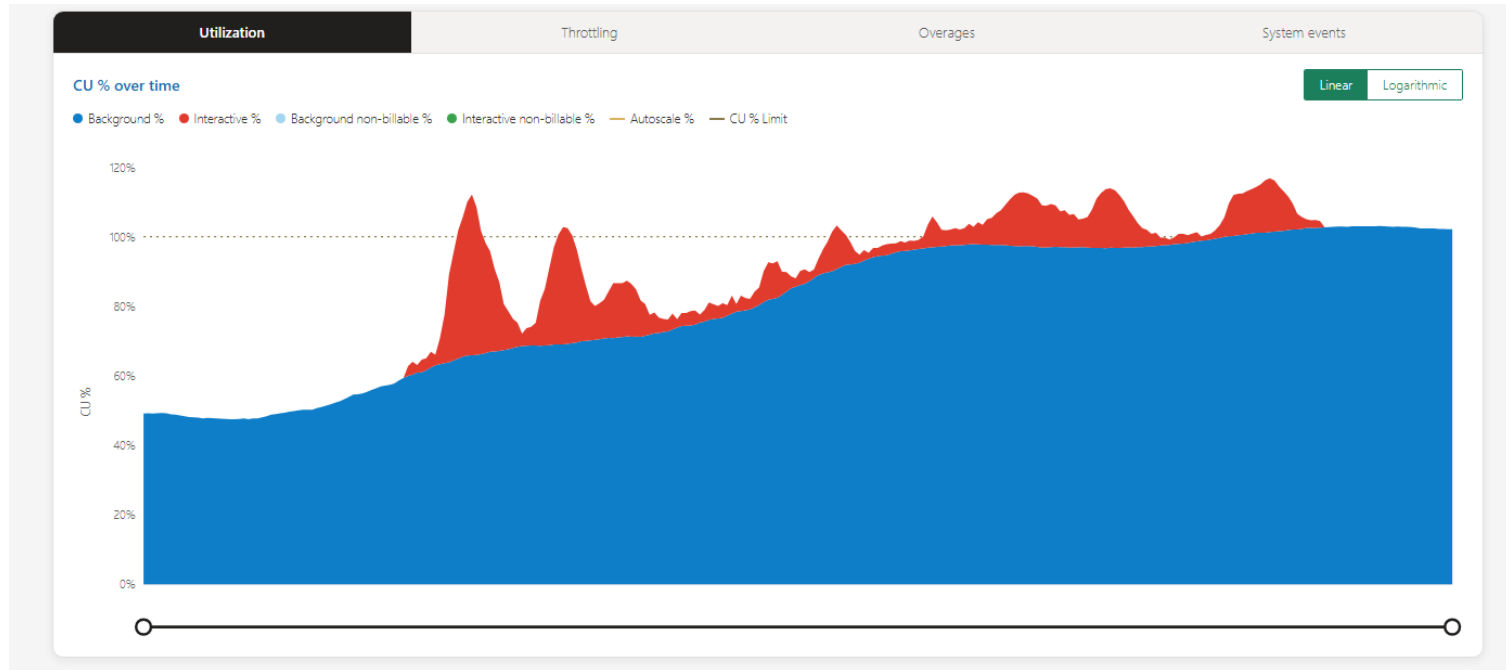
Large user base, DirectQuery heavy visuals, large reports, paginated reports, subscriptions ... too much in parallel.

Mitigation

- look to optimize reports or datasets
- capacity overage billing
- are Power BI workspaces an option?

Sustained overage

24hr pattern



Symptoms

Background alone eats the capacity and carries further than the 24h threshold.

Possible reasons

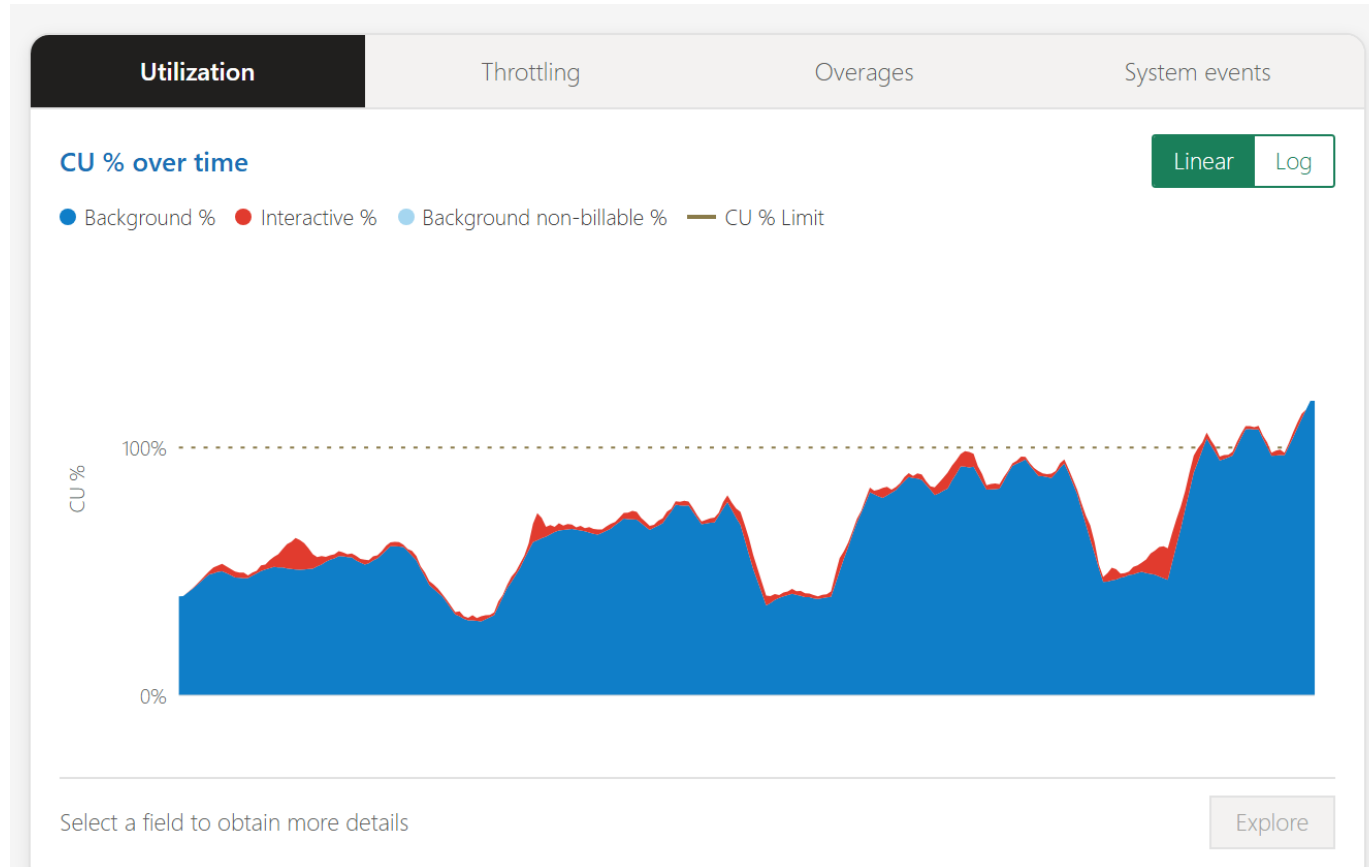
The capacity is undersized for the workloads. One job is hardly inefficient and never got fixed. A heavy model or query is building up background activities.

Mitigation

- reduce frequency
- leverage incremental refresh
- scale up

Adoption creep

monthly pattern



Symptoms

The baseline drifts up a few percent reveryday. At the end of the month you're over capacity. No single day explains it.

Possible reasons

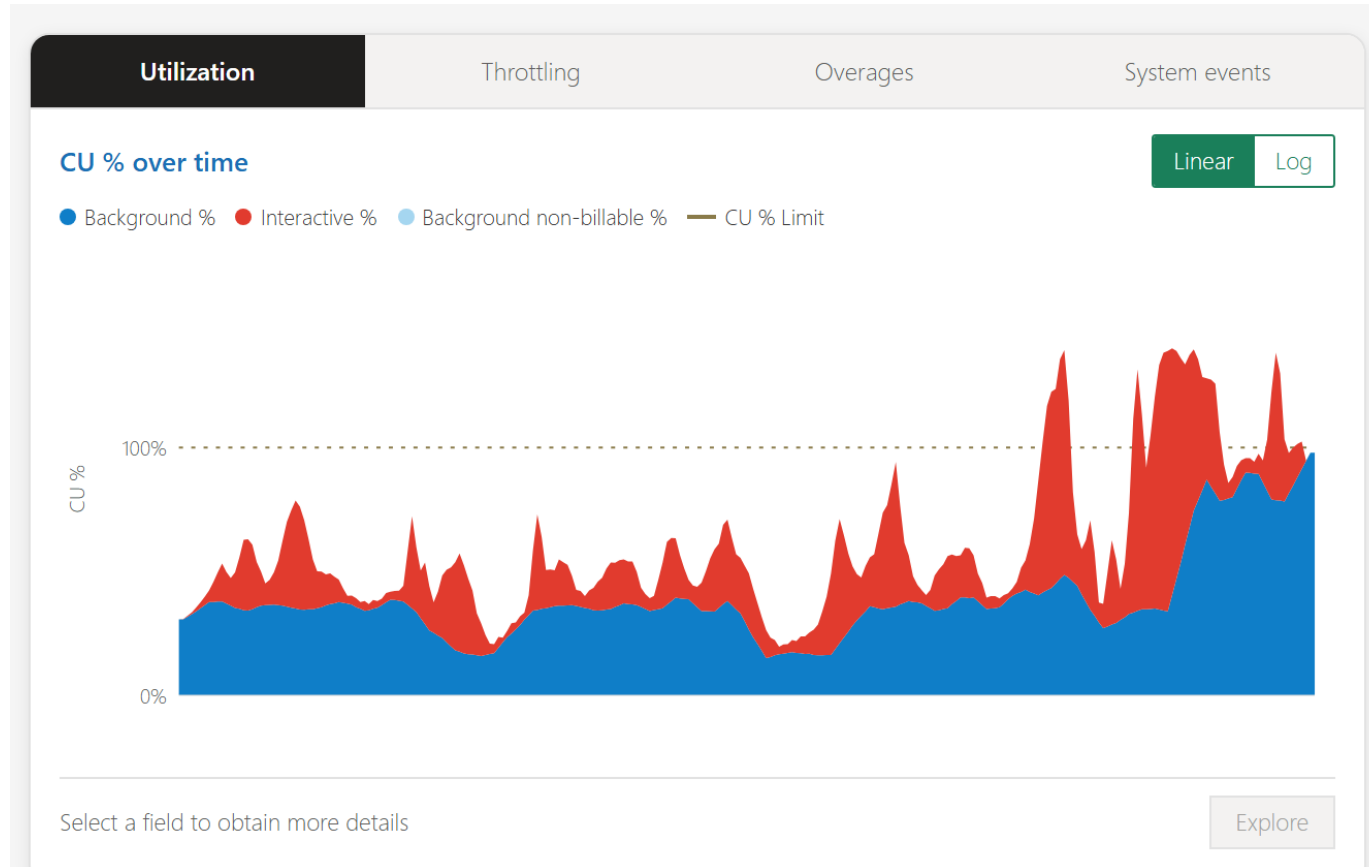
Adoption (more reports, more users, more refreshes, ...)
Typically no single reason or workload is the cause

Mitigation

- scaling or adding new capacity

Month-end close

monthly pattern



Symptoms

A normal month, then a peak past the limit for several days in a row.
The overage sits inside a specific window.

Possible reasons

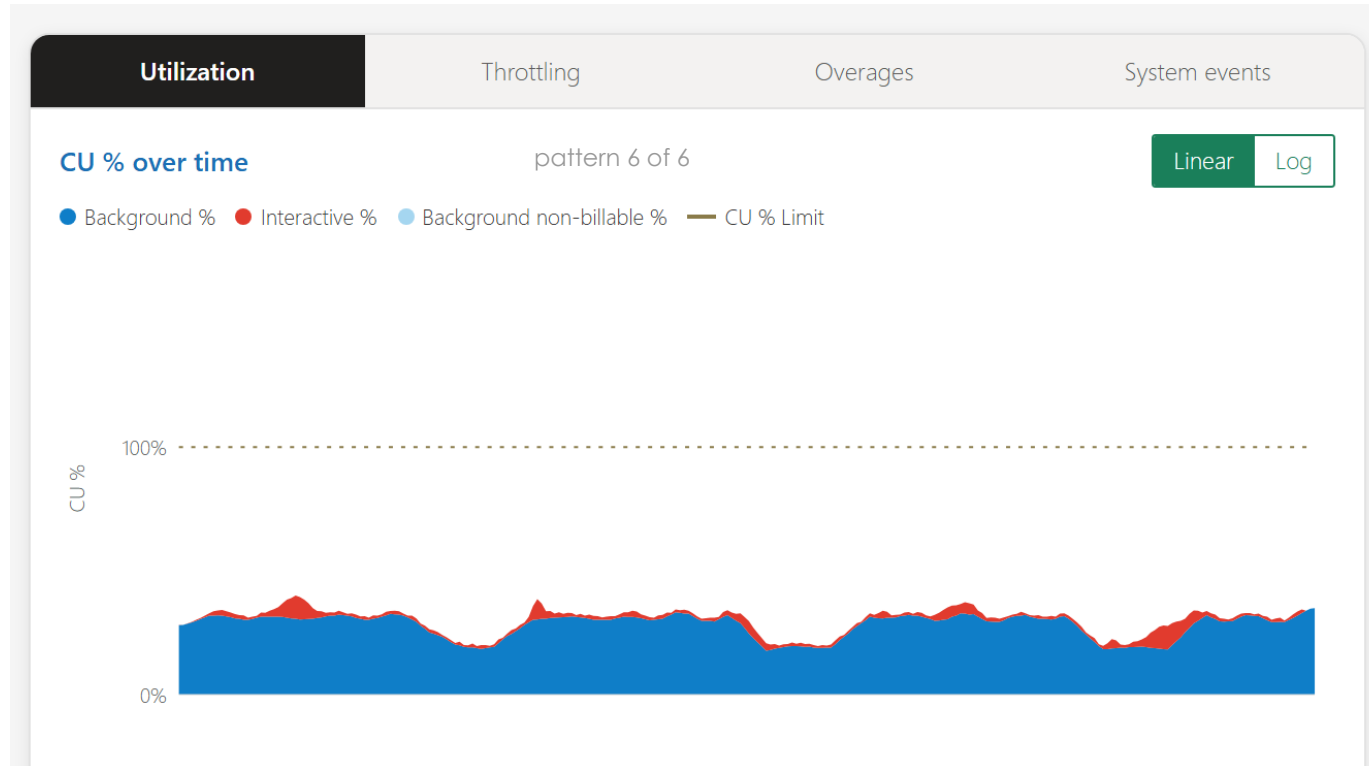
Month-end consolidation, allocations and reporting land in the same short window, causing increased load.
Typically more refreshes & users throughout these closing days.

Mitigation

- if only sporadic overage happens – capacity overage billing
- otherwise: scale up (during closing periods)

Consistently under-used

monthly pattern



Symptoms

A flat, low plateau, small peaks reaching only about half capacity, and a permanent gap to the ceiling.

Possible reasons

The SKU was sized for a peak or a project that turns out to be smaller, a workload moved off or stopped.
Initial sizing was incorrect

Mitigation

Scale down + reserve to save money

Optimise costs

the capacities

- extra reservations
- downscaling or pausing where possible

the storage

Move data to cool or cold OneLake Storage

Hot
€ 0,0197

Cool
€ 0,0107

Cold
€ 0,0034

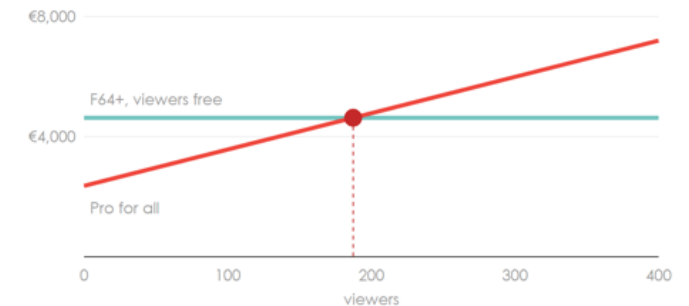
North Europe, per GB per month.

the licenses

Pro or PPU near threshold?

Pro 1 GB per semantic model

PPU 100 GB per semantic model



- Azure **Cost Management + Fabric Cost Analysis** reports Euros
- The **capacity metrics app** and/or **FUAM** reports utilization
- **Microsoft 365 admin center** shows Power BI licences

Optimise decision tree

1

mitigate

- Fix the low-hanging fruit heavy CU hogs
- Surge protection

2

scale

- Scale up (reserve)
- Capacity overage
- Pausing (bill shock-

3

isolate

- Copilot capacity
- Spark Autoscale billing
- New capacity

Optimise costs

Closing the cycle

1 plan

- Choose t-shirt sized capacity
- Check licence needs
- Plan capacities based on workspace strategy

2 monitor

use a combination of tools



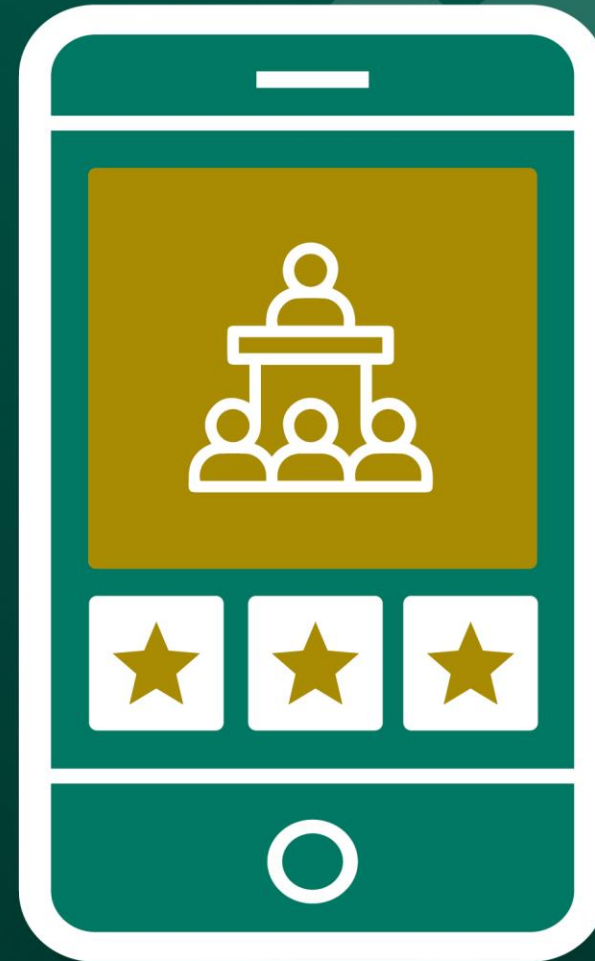
identify patterns

3 optimise

-  Mitigate
-  Scale
-  Isolate



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